Ground-Truth-First Benchmarking of Reinforcement Learning via Converse Optimality

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Abstract

Benchmarking in reinforcement learning (RL) is usually relative: algorithms are ranked by returns on environments whose optimal policies are unknown. Distance to optimality is unobservable, and empirical conclusions can be entangled with reward design, stochasticity, hyperparameter choices, environment selection, and simulator details. We develop a ground-truth-first benchmarking framework based on converse optimality. The paper unifies two complementary modes. Sensitivity benchmarking samples a family of certified environments from a continuous-time generator and tests whether a learning pipeline remains consistent under task variation; oracle-referenced comparison uses native discrete-time generators and evaluates learned policies against the exact optimum via optimality gap, regret, and policy error. The results also separate the roles of time formalism. Continuous-time generators are natural for physically interpretable sensitivity studies, but their realized behavior depends on the simulator step Δt and numerical scheme. Native discrete-time generators remove this additional degree of freedom: the environment, oracle, and learner operate on the same transition law, so evaluation is not confounded by an external integration choice. We make the manuscript empirically self-contained by reproducing representative quantitative results from both settings, including optimizer sensitivity and oracle-referenced comparisons on serial planar arms and dynamically extended nonholonomic vehicles. A public project page hosts code, fixtures, and diagnostics. The methodology extends beyond smooth affine Gaussian examples: non-affine action interfaces, finite-covariance non-Gaussian noise, and non-smooth value functions can all be incorporated.

1 Introduction

Performance evaluation has become the default empirical language of reinforcement learning. A new method is introduced, evaluated on benchmark environments, and compared against a changing set of baselines. This protocol is valuable, but it leaves a basic control-theoretic quantity hidden: *distance to optimality*. When the optimal policy is unknown, a benchmark can tell us that one algorithm achieved a larger return than another; it cannot tell us whether that algorithm is near optimal, whether all compared methods are poor, or whether the ranking is an artifact of reward scaling, random seeds, task weighting, environment selection, or simulator conventions.

This limitation is not superficial; it is fundamentally structural. Deep RL conclusions are well-known to depend strongly on implementation choices, evaluation protocols, random seeds, hyperparameters, and statistical reporting practices [1–3]. For example, aggregate rankings may fluctuate significantly when tasks are weighted or normalized differently [4]. Recent critiques therefore argue that benchmark scores alone are insufficient evidence for broad algorithmic claims; controlled scientific tests are needed to explain how and when algorithms work [5]. For RL control, two particular questions arise: *does a learning pipeline remain reliable when the environment varies?* and *how far is a learned*

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policy from the true optimum on a fixed task? Conventional benchmarks struggle with both when the optimum is unavailable.

Converse optimality offers a direct alternative. It reverses the usual direction of optimal control: rather than solving a Bellman or Hamilton–Jacobi–Bellman equation, one starts from a desired value function and policy and constructs environments for which they are optimal. The environment is therefore generated together with its certificate, turning evaluation from relative scoring to absolute calibrated measurement: optimality gaps, regret, Bellman residuals, and sensitivity across distributions of certified tasks.

The central claim of this paper is that converse optimality transforms RL benchmarks into *absolute known-answer experimental instruments*. A score-based benchmark mainly asks which method obtains the highest return. A ground-truth-first benchmark grounded in converse optimality can also ask why a method fails, how far it is from optimal, whether conclusions survive controlled environment variation, and whether the strongest baseline is genuinely close to the certified optimum.

We organise this view into two complementary modes. In *sensitivity benchmarking*, a generator defines a distribution over certified environments, turning environment variation into an experimental variable that tests pipeline consistency. In *oracle-referenced comparison*, a fixed fixture provides a certified pair $(V^\star, \pi^\star)$, enabling absolute evaluation by optimality gap, regret, and policy error.

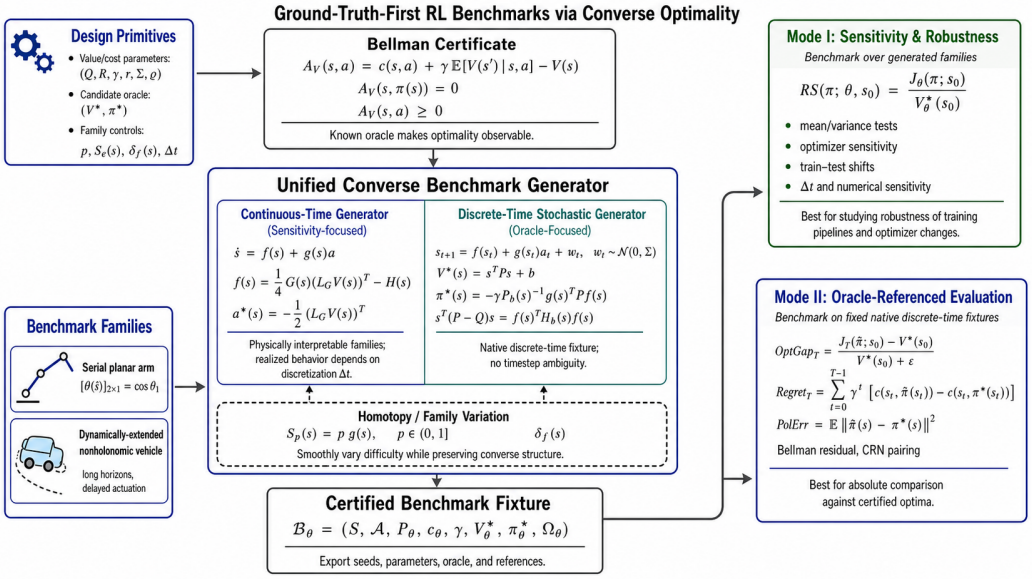


Figure 1: Ground-truth-first benchmark-generation pipeline. A converse construction samples certified dynamics and costs, validates the optimality certificate, exports an environment fixture, trains learners without oracle access, and evaluates learned policies against the hidden pair $(V^\star, \pi^\star)$.

Time formalism is part of this design. Continuous-time generators produce physically interpretable non-linear systems, but the MDP experienced by a digital learner depends on the integration step Δt and numerical scheme. This dependence is not a defect; it makes continuous-time generators well-suited for sensitivity studies. Native discrete-time generators define the transition law, oracle, and evaluation protocol in the same formalism, removing the discretisation confound. Thus, continuous-time generators are natural for sensitivity, whereas native discrete-time fixtures are natural for absolute algorithm comparison.

Our contribution is methodological. We introduce a benchmark design principle in which optimality is built into the environment specification and hidden until evaluation. Concretely, we contribute: (1) a ground-truth-first fixture formalism exporting certified values and policies; (2) a separation between sensitivity studies over generated families and oracle-referenced comparison on fixed fixtures; (3) a time-formalism analysis; (4) a validation and evaluation protocol using Bellman consistency, closed-loop checks, and common-random-number schedules; and (5) an empirical synthesis demonstrating

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optimizer sensitivity in continuous-time systems and absolute evaluation in native discrete-time robotic control families.

The accompanying anonymous project page provides the manuscript, code, fixtures, results, and diagnostics: `https://gtfrl.github.io/Benchmarking-of-Reinforcement-Learning-via-Converse-Optimality/`. The studied systems are examples of a broader generator principle, not a closed catalogue. The same viewpoint can incorporate non-affine raw actions, finite-covariance non-Gaussian disturbances, partial or image-valued observations over certified latent dynamics, and non-smooth value functions.

2 Related Work

RL benchmark suites. Shared environment suites—ALE, MuJoCo, Gym, Procgen, MiniGrid, bsuite, D4RL [6–12]—enable large-scale empirical comparison but generally lack known optimal policies. Their primary signal is relative return, and a method may outperform baselines while remaining far from optimal.

Empirical reliability. RL conclusions depend heavily on implementation details, random seeds, and statistical reporting [1–3]. Benchmark scores alone are insufficient evidence for broad claims [5]. We adopt this diagnostic perspective but change the experimental substrate: generating tasks with certified optima makes absolute error observable.

Converse optimality. Converse optimality constructs systems and costs for which a prescribed value or policy is optimal, using HJB certificates in continuous time [13] and Bellman-equation conditions in discrete time [14]. It has been used to generate deterministic RL benchmarks and study sensitivity over environment families [15], and extended to stochastic control-affine fixtures with direct oracle metrics [16]. We synthesize these lines into a single benchmarking framework.

Generality. Smooth, affine, Gaussian choices give compact formulas but are not essential. Non-smooth value functions can be handled via viscosity solutions and Clarke subdifferentials [17–20]; in quadratic-value settings, additive noise need not be Gaussian because the Bellman expectation depends only on the disturbance covariance. We include these extensions to clarify scope while keeping the main experiments focused on explicit oracles.

3 Ground-Truth-First Fixtures

A standard RL benchmark provides an environment but not the optimal return. A *ground-truth-first* benchmark adds a certificate that a particular value function $V^\star$ and policy $\pi^\star$ are indeed optimal. We work within the usual Markov decision process (MDP) framework with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel P, stage cost c, and discount γ. For a stationary policy π, the value function is

$$V^\pi(s) = \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k c(s_k, \pi(s_k)) \;\middle|\; s_0 = s\right],$$

and the optimal value function satisfies $V^\star(s) = \inf_\pi V^\pi(s)$.

To check optimality, we use the *Bellman residual* $\mathcal{R}_V(s,a) = c(s,a) + \gamma \int V(s')\, P(\mathrm{d}s' \mid s,a) - V(s)$. The *optimality certificate* states: if for all states and actions $\mathcal{R}_V(s,\pi(s)) = 0$ and $\mathcal{R}_V(s,a) \geq 0$, then $V = V^\star$ and π is optimal (see Appendix C).

A *benchmark fixture* $\mathfrak{B}_\theta = (\mathcal{S}, \mathcal{A}, P_\theta, c_\theta, \gamma, V_\theta^\star, \pi_\theta^\star, \Omega_\theta)$ packages the certified MDP with a validation record. The agent trains on the MDP $(\mathcal{S}, \mathcal{A}, P_\theta, c_\theta, \gamma)$; $V_\theta^\star$ and $\pi_\theta^\star$ are withheld as evaluation oracles. When θ is fixed, the fixture provides a single known-answer task. When θ is sampled from a distribution, one obtains a *distribution over certified MDPs*, enabling the two modes in Table 1.

4 Mode I – Continuous-time sensitivity benchmarking

The abstract certificate of Section 3 guarantees optimality when certain conditions hold. To turn that certificate into a usable benchmark, we need *converse generator templates*—recipes that, given

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Table 1: Two complementary benchmark modes enabled by ground-truth-first fixtures.

Mode	Scientific question	Typical outputs
Sensitivity / robustness	Does a learning pipeline remain reliable when the environment varies?	Relative suboptimality distributions, mean and variance tests, train–test shift studies
Oracle-referenced comparison	How far is a learned policy from the certified optimum on a fixed fixture?	Optimality gap, regret, Bellman residual, policy error, paired CRN confidence intervals

a desired value function V and policy, produce the dynamics and costs that satisfy the certificate. Mode I uses a *continuous-time* generator. Its main strength is that it produces physically interpretable, non-linear control systems, making *environment variation* a controlled experimental variable.

4.1 Continuous-time converse generators for sensitivity

Consider a continuous-time, control-affine system $\dot{s} = f(s) + g(s)a$ with state $s \in \mathbb{R}^n$ and action $a \in \mathbb{R}^m$. We prescribe a positive definite C^1 value function V. A converse construction [15] uses two designer-chosen smooth fields: a vector field $H(s)$ shaping the drift, and a matrix field $G(s)$ directing the control. The generator outputs

$$g(s) = G(s), \quad f(s) = \frac{1}{4}G(s)\big(\mathcal{L}_G V(s)\big)^\top - H(s), \quad c(s,a) = \mathcal{L}_H V(s) + \|a\|_2^2,$$

where $\mathcal{L}_G V(s)$ is the Lie derivative of V along the columns of G. A completion of squares (Appendix D.1) gives the Hamiltonian as $\|a + \frac{1}{2}(\mathcal{L}_G V(s))^\top\|_2^2$, yielding the certified optimal policy $a^\star(s) = -\frac{1}{2}(\mathcal{L}_G V(s))^\top$.

From generator to sensitivity benchmark. The fields H and G are *knobs* that parametrize a rich family of certified nonlinear control problems (stable/unstable, underactuated/fully actuated, stiff/slack). Sampling them from a distribution produces a *distribution over certified MDPs*—the substrate for sensitivity and robustness studies. Because the certificate is continuous-time, the digital RL agent experiences a discretised version whose dynamics depend on the step size Δt and integration method. This dependency is not a bug: it makes environment variation and numerical realization part of the experimental variable, precisely what sensitivity studies require.

4.2 Sensitivity evaluation protocol

A continuous-time generator defines a distribution over certified environments indexed by θ. For a learning procedure $\mathcal{L}$ and a sampled environment θ, we obtain a trained policy π_θ. The *relative suboptimality* (RS) normalises the incurred cost by the known optimal cost:

$$\mathrm{RS}(\pi_\theta; \theta, s_0) = \frac{J_\theta(\pi_\theta; s_0)}{V_\theta^\star(s_0) + \epsilon},$$

where J_θ is the cumulative cost and ϵ a small constant to avoid division by 0. RS is task-aware: it measures excess cost as a multiple of the optimum, independent of raw task scale.

Sampling N environments and training on each yields a sample $\{\mathrm{RS}_i\}$. Two procedures are compared by their RS distributions—the question is not who wins on one task, but whether one procedure systematically produces a different suboptimality profile across tasks. We apply an F-test for variance (consistency) and a t-test for means (average performance). The protocol is: sample environments, train, compute RS with the known $V_\theta^\star$, and test. This turns environment variation from nuisance into a controlled, statistically measurable experimental variable.

4.3 Sensitivity experiment

We illustrate the sensitivity protocol with the continuous-time case study of a generator producing a sample of 160 unstable three-dimensional control problems, each with a known optimal value [15]. Two PPO pipelines, identical in all respects except the optimizer (Adam *vs.* RAdam), were trained on independent splits of 80 environments.

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Relative suboptimality is computed for each trained policy π_θ. An F-test comparing the two RS distributions gave a variance ratio $F \approx 2.59$ ($p \approx 1.7 \times 10^{-5}$), rejecting the null of equal variance at any conventional significance level.

This result does not mean one optimizer is uniformly "better." It shows that the optimizer choice materially alters the *spread* of suboptimality across environments—a measure of pipeline *consistency*. A return-only benchmark would have obscured this effect entirely. The converse-generated family makes sensitivity a measurable and statistically testable property.

Table 2: Continuous-time sensitivity case study. Two PPO pipelines differed only in optimizer. The measured quantity is relative suboptimality over a generated family of certified environments.

Quantity	Value
Generator family	Unstable 3D continuous-time converse-optimal systems
Learning algorithm	PPO [21]
Perturbation under test	Adam vs. RAdam optimizer
Environment sample size	160 generated problems, split into 80 per optimizer arm
Normalization	Relative suboptimality using each environment's known optimal value
Test statistic	Variance ratio $F \approx 2.59$
p-value	$\approx 1.7 \times 10^{-5}$
Interpretation	Optimizer choice materially changed outcome variance across certified environments

5 Mode II – Native discrete-time oracle-referenced comparison

We now present converse generator templates that operate directly in discrete time. We proceed from deterministic dynamics [14], to stochastic dynamics with Gaussian noise, and then to extensions with non-affine action interfaces and non-Gaussian disturbances. Throughout, the generator produces an MDP together with a certified optimal value-policy pair, exactly as required by Proposition 1 (Appendix C).

5.1 Deterministic discrete-time generator

Consider a deterministic, control-affine, discrete-time system $s_{k+1} = f(s_k) + g(s_k)\, a_k$, with state $s \in \mathbb{R}^n$ and action $a \in \mathbb{R}^m$. The stage cost $c(s,a) = q(s) + \rho(a)$ is separable, with $q \geq 0$ and ρ is strictly convex with $\rho(0) = 0$. For a given input-coupling function g and control cost ρ, the converse optimality problem asks: given a candidate optimal value function $V^\star$, find a drift f such that $V^\star$ satisfies the Bellman equation for the considered system.

Göhrt *et al.* [14] provided a complete answer. A pair $(V^\star, q)$ is *admissible*—i.e., there exists a drift f realizing $V^\star$ as the optimal value function—if and only if $V^\star(s) - q(s) \geq 0$ for all $s \in \mathbb{R}^n$. The sufficiency proof is constructive and uses a homotopy argument (see Appendix D). When the value function, the state cost, and the control cost are quadratic, with $V^\star(s) = s^\top P s$, $q(s) = s^\top Q s$, $\rho(a) = a^\top R a$ and $P, Q, R \succ 0$, the admissible drift can be written explicitly as

$$f(s) = H(s)^{-1/2}\, S(s)\, (P - Q)^{1/2}\, s, \qquad S(s)^\top S(s) = I,$$

where $H(s) = \left(P^{-1} + g(s) R^{-1} g(s)^\top\right)^{-1}$. The orthogonal field $S(s)$ can be chosen freely and serves as a design knob for injecting nonlinear coupling while preserving optimality. The optimal policy is $a^\star(s) = -\left(R + g(s)^\top P g(s)\right)^{-1} g(s)^\top P f(s)$.

5.2 Stochastic discrete-time generator

The converse framework can also be extended to stochastic systems $s_{k+1} = f(s_k) + g(s_k) a_k + w_k$ with additive Gaussian noise $w_k \sim \mathcal{N}(0, \Sigma)$ [16]. The optimal value function satisfies the stochastic Bellman equation

$$V^\star(s) = \inf_a \{c(s,a) + \gamma \mathbb{E}[V^\star(f(s) + g(s)a + w)]\}.$$

An analogous necessary-and-sufficient optimality condition holds (see Appendix C).

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For benchmark generation, the most useful instance is the *quadratic-Gaussian* case with value function $V^\star(s) = s^\top P s + b$, noise certainty equivalence offset b and quadratic stage costs. The stochastic Bellman equation reduces to a single *energy equation*

$$s^\top (P - Q)s = f(s)^\top H_\gamma(s) f(s),$$

with an explicit state-dependent metric $H_\gamma \succ 0$, and which any drift must satisfy to make $V^\star$ optimal. Two explicit parameterizations exist: a *metric-normalized* form $f(s) = H_\gamma^{-1/2} S(s)(P - Q)^{1/2} s$ with an orthogonal field $S(s)$, and a *square-root* form. Both inject bounded nonlinear coupling while preserving the energy identity exactly. A control-strength homotopy $g_p(s) = p\, g(s),\ p \in (0, 1]$ varies authority, conditioning, and task difficulty without invalidating the certificate.

In native discrete time, the transition law itself is the benchmark: learner, oracle, and metrics share the same map. This eliminates discretisation confounds and makes these fixtures ideal for absolute, calibrated algorithm comparison (Mode II).

5.3 Example families

The following families instantiate a general recipe rather than exhaust it. Each family highlights a particular design dimension such as state-dependent actuation, dynamic extension, non-affine interfaces, or non-smooth value structure that the generator can accommodate. Complexity is introduced through certified parameters: link count, module count, dynamic-extension depth, coupling strength, control authority, actuator coordinates, disturbance structure, and observation maps. A certified latent MDP can also be wrapped with partial or image-valued observations while preserving the oracle on the hidden state.

State-dependent actuation and nonlinear coupling. As a concrete example, we consider a first family which models a serial n-link planar arm with state $s = [\theta_1, \omega_1, \ldots, \theta_n, \omega_n]^\top \in \mathbb{R}^{2n}$ and action $a = [\tau_1, \ldots, \tau_n]^\top \in \mathbb{R}^n$. The input map is configuration dependent with $[g(s)]_{2i,i} = \cos\theta_i$ for $i = 1, \ldots, n$. with all other entries zero. The drift is generated by

$$f_p(s) = H_\gamma^{(p)}(s)^{-1/2} S_p(s)(P - Q)^{1/2} s. \tag{1}$$

The orthogonal field $S_p(s)$ is a product of Givens rotations containing $\tanh$ terms that create bounded velocity-dependent and neighbor-coupling effects. A sinusoidal term gives a gravity-like bias. The family is non-linear, scalable in dimension, and certified by construction.

Dynamic extension and delayed actuation. The second example of benchmark family illustrates dynamic extension with a nonholonomic vehicle family [16] where for module i, the state is $s_i = [x_i, y_i, \phi_i, v_i^{(0)}, \ldots, v_i^{(r_v-1)}, \omega_i^{(0)}, \ldots, \omega_i^{(r_\omega-1)}]^\top$ and the action is $a_i = [a_{v,i}, a_{\omega,i}]^\top$. The kinematic states evolve as

$$x_{i,k+1} = x_{i,k} + \tau v_{i,k}^{(0)} \cos\phi_{i,k}, \quad y_{i,k+1} = y_{i,k} + \tau v_{i,k}^{(0)} \sin\phi_{i,k}, \quad \phi_{i,k+1} = \phi_{i,k} + \tau \omega_{i,k}^{(0)}.$$

Velocity chains shift upward, with control entering only at the highest derivative. We stack modules through the matrix block diagonal matrix G, and we obtain the overall state of the system $s_{k+1} = f_p(s_k) + pGu_k + w_k$. Dynamic extension introduces delayed actuation, long credit-assignment paths, and stiff closed-loop behavior, stressing RL algorithms differently from the arm family.

5.4 Oracle-referenced evaluation protocol

Fixtures are validated (see Appendix F) before export. We measure four absolute metrics.

Normalized optimality gap (tail-corrected): the tail-corrected cost gap is

$$\mathrm{Gap}_T^c(\hat{\pi}; s_0) = \frac{\mathbb{E}\Big[\sum_{k=0}^{T-1} \gamma^k c(s_k, \hat{\pi}(s_k)) + \gamma^T V^\star(s_T)\Big] - V^\star(s_0)}{|V^\star(s_0)| + \epsilon}.$$

where $\epsilon > 0$ guards against division by zero. The tail term removes truncation bias, yielding zero for the optimal policy (up to numerical precision) and nonnegative values for suboptimal policies. Under reward convention, the signed gap is $G^\star = -V^\star$, so zero is optimal and more negative values indicate larger suboptimality. The paper clearly states whether Gap_T^c or Gap_T^r is used in each plot.

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Discounted regret: Regret accumulates the per-step excess cost incurred by $\hat{\pi}$ relative to the optimal policy. It is a trajectory-level diagnostic that complements the scalar gap. Lower cost regret is better.

$$\mathrm{Regret}_T^c(\hat{\pi}; s_0) = \mathbb{E}\left[\sum_{k=0}^{T-1} \gamma^k \Big(c(s_k, \hat{\pi}(s_k)) - c(s_k, \pi^\star(s_k)) \Big)\right].$$

If a reward-convention regret is plotted, it is defined by $\mathrm{Regret}_T^r = -\mathrm{Regret}_T^c$, and the sign convention is stated in the corresponding caption.

Policy error: $\mathbb{E}_{s\sim\mu}\|\hat{\pi}(s) - \pi^\star(s)\|_2^2$. **Bellman residual**: $\mathcal{R}_{\hat{V}}$ if a critic $\hat{V}$ is available.

Common Random Numbers (CRN): All policies are evaluated on a shared pre-generated schedule of start states and noise streams. Paired bootstrap over trajectories yields confidence intervals, ensuring that differences reflect algorithms, not noise realisations.

5.5 Oracle-referenced experiment

We evaluate PPO, A2C, SAC, TD3, and DDPG on ConverseArm and NVDEx native discrete-time fixtures under a common-random-number schedule, using both diverse initial states (`serial_random`) and a fixed initial state (`serial_fixed`); NVDEx adds a longer horizon ($T = 700$) with delayed actuation. Figure 5 shows that the oracle-referenced ranking is strongly regime-dependent. On the serial-random fixtures, TD3 has the best point estimate in both signed gap and regret, followed by SAC and PPO; A2C and DDPG remain substantially farther from the certified optimum. This suggests that the fixed-initial-state regime may favor the stability of the reproduced PPO protocol.

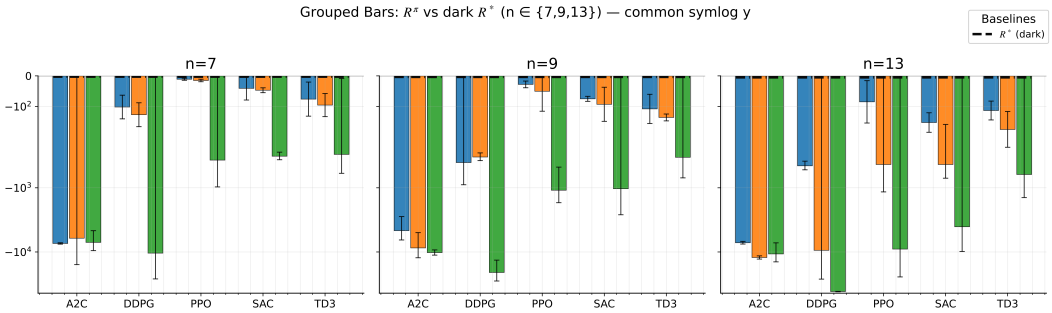


Figure 2: Comparing algorithm absolute performance. Discounted reward bars ($r = -c$) with 95% CIs under CRN. Colors correspond to $p \in \{0.5, 0.6, 0.8\}$.

The ranking changes on the serial-fixed fixtures. PPO is closest to the oracle by a large margin, while SAC and TD3 exhibit large regret despite their strong performance in many standard continuous-control benchmarks. This indicates that the fixed-initial-state regime creates a narrow, stationary occupancy distribution in which PPO's clipped on-policy updates are more stable, whereas off-policy bootstrapping and replay do not provide the same advantage.

The NVDEx regime shows the strongest separation: PPO is near the certified optimum, while DDPG, SAC, TD3, and A2C remain far from zero. This reflects the vehicle's dynamic extension: delayed actuation, longer horizon, and error propagation through feedback-sensitive dynamics. The result does not claim universal PPO superiority; rather, it shows that this fixture exposes a failure mode where delayed actuation and long-horizon conditioning challenge off-policy baselines under the reproduced protocol.

Together, Figures 5 and Appendix F.3 show that certified fixtures turn RL evaluation from relative ranking into diagnostic measurement. No single algorithm dominates: TD3 excels on serial-random, while PPO excels on serial-fixed and NVDEx. Distributional plots expose lower-tail failures hidden by means. Because the optimum is known, we can measure not only which method performs best, but also how far each is from optimality and how error is distributed across runs and fixtures.

In the serial-random regime, A2C is concentrated far below zero, while DDPG has a wide distribution with a severe negative tail. PPO, SAC, and TD3 are shifted closer to the oracle, with TD3 having the best central performance. This supports the interpretation that randomized initial states favor algorithms that can exploit broad transition coverage, while still revealing instability in DDPG.

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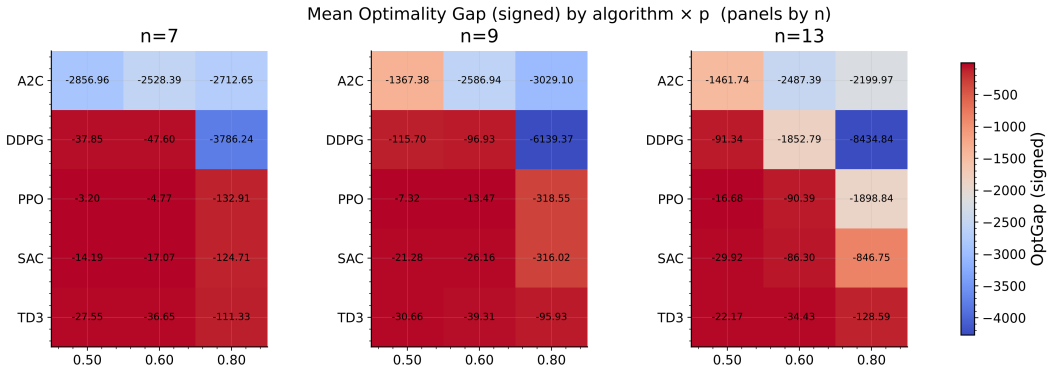


Figure 3: Serial n-link planar arm (ConverseArm): absolute oracle-referenced performance (optimality gap) across algorithms (rows) and control authority p (columns), with separate panels for n.

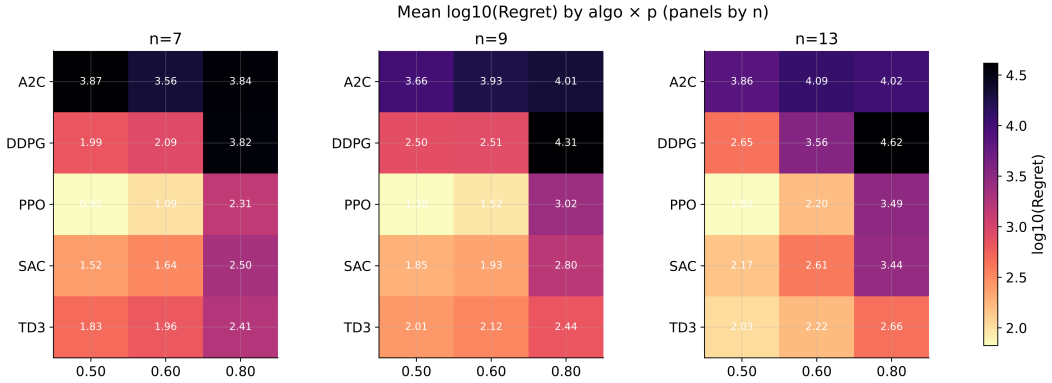


Figure 4: Serial n-link planar arm (ConverseArm): absolute oraclereferenced performance (regret) across algorithms (rows) and control authority p (columns), with separate panels for n.

In the serial-fixed regime, PPO's distribution is shifted closest to zero, whereas SAC and TD3 exhibit lower-tail failures. Thus, the performance change from serial-random to serial-fixed is not only a change in mean score; it is a change in the shape of the error distribution. The fixed regime appears to favor stable on-policy improvement over replay-based reuse.

The NVDEx panel is the clearest example of why oracle-referenced distributions are useful. PPO remains concentrated near the optimum, while the other baselines show broad distributions and severe negative tails. Since NVDEx contains dynamic extension and delayed actuation, these tails indicate that small policy or value-estimation errors can be amplified over long horizons. The certified value function makes this failure severity measurable in oracle units.

6 Extensions beyond the basic fixtures

6.1 Non-affine system under non-Gaussian noise

The fixture methodology is not limited to affine actions or Gaussian disturbances. As a concrete illustration, we present a planar aerial robot family where the RL agent controls raw motor commands through a non-linear actuator map (a component-wise cubic diffeomorphism), while the certified optimum is constructed in a latent affine coordinate. The resulting environment is non-affine in the raw action, yet the oracle remains analytically exact. In the quadratic value setting, the optimal policy depends on the disturbance only through its covariance, so any zero-mean finite-covariance noise—Gaussian or heavy-tailed—preserves the same optimum. The full construction is given in Appendix E. These extensions underscore the generality of the oracle-fixture viewpoint; experiments in this paper focus on affine, Gaussian families with explicit oracles and cheapest validation.

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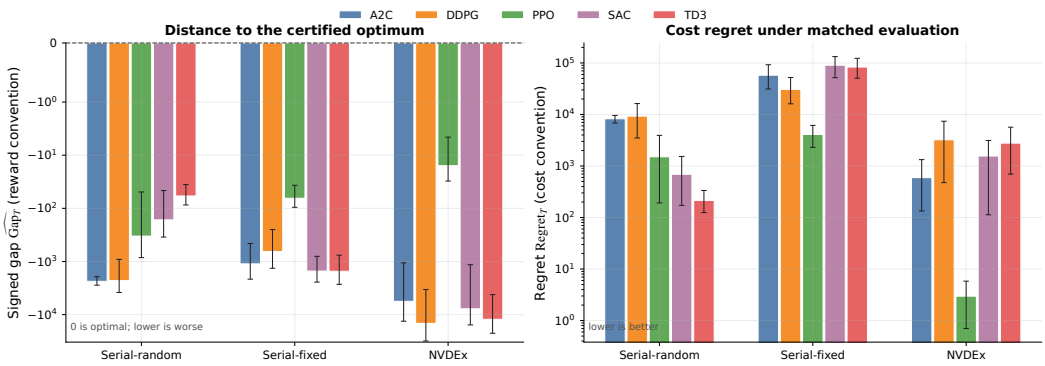


Figure 5: Left: reward-convention signed gap to the certified optimum, where zero is optimal and more negative values are worse. Right: cost regret under matched CRN evaluation, where lower is better. Bars show means and error bars show paired-bootstrap 95% confidence intervals. The logarithmic vertical scales show order-of-magnitude differences across algorithms and regimes.

6.2 Non-smooth value functions

The fixture viewpoint extends to value functions that are merely locally Lipschitz, such as piecewise-quadratic templates with kinks along switching surfaces. In continuous-time deterministic control, classical gradients are replaced by Clarke subdifferentials, and the smooth HJB certificate is replaced by a viscosity-HJB inequality. A homotopy parameter selects a generalized gradient branch on the switching surface, generating families of benchmarks with tunable non-smoothness and discontinuous optimal policies. The full construction of continuous-time converse optimality with non-smooth value functions, including a worked two-dimensional piecewise-quadratic family, is given in Appendix G. The discrete-time non-smooth extension is currently under development.

7 Discussion

The framework trades broad ecological coverage for calibration: converse-generated fixtures provide known-answer control problems with measurable optimality gaps, not replacements for open-ended simulators or real-world platforms. The families studied here exhibit nonlinear drift, state-dependent actuation, delayed input channels, and stiff closed-loop behaviour, with difficulty tunable via dimension, dynamic extension, coupling strength, control authority, and disturbance structure. They remain constructed tasks; strong performance indicates diagnostic capability under certified dynamics, not deployment readiness.

Open challenges include temporally correlated disturbances, hard state/input constraints, partial observability, multi-agent settings, and oracle preservation under representation learning. A methodological risk is overfitting to the generator distribution; reports should therefore specify generator parameters, validation domains, and rejection criteria.

8 Conclusion

We presented a ground-truth-first framework for RL benchmarking based on converse optimality. The central idea is to make the optimum part of the benchmark specification rather than an unknown target outside it. This supports two complementary modes: generated families for sensitivity and robustness studies, and fixed known-answer fixtures for oracle-referenced comparison by optimality gap, regret, Bellman residual, and policy error. Continuous-time and discrete-time generators play distinct roles: the former provide physically interpretable sensitivity probes but inherit simulator-step dependence, while the latter remove timestep ambiguity for absolute comparison. The studied systems are examples of a broader methodology. By extending the same certificate logic to non-affine raw actions, finite-covariance non-Gaussian disturbances, and non-smooth value functions, converse optimality enables RL benchmarks that do not merely rank policies but measure their distance from known optimal behavior.

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References

[1] Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.

[2] Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.

[3] Andrew Patterson, Samuel Neumann, Martha White, and Adam White. Empirical design in reinforcement learning. *Journal of Machine Learning Research*, 25(318):1–63, 2024.

[4] David Balduzzi, Karl Tuyls, Julien Perolat, and Thore Graepel. Re-evaluating evaluation. *Advances in Neural Information Processing Systems*, 31, 2018.

[5] Scott M Jordan, Adam White, Bruno Castro Da Silva, Martha White, and Philip S Thomas. Position: Benchmarking is limited in reinforcement learning research. *arXiv preprint arXiv:2406.16241*, 2024.

[6] Marc G Bellemare, Yavar Naddaf, Joel Veness, and Michael Bowling. The arcade learning environment: An evaluation platform for general agents. *Journal of artificial intelligence research*, 47:253–279, 2013.

[7] Emanuel Todorov, Tom Erez, and Yuval Tassa. Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ international conference on intelligent robots and systems*, pages 5026–5033. IEEE, 2012.

[8] Greg Brockman, Vicki Cheung, Ludwig Pettersson, Jonas Schneider, John Schulman, Jie Tang, and Wojciech Zaremba. Openai gym. *arXiv preprint arXiv:1606.01540*, 2016.

[9] Karl Cobbe, Chris Hesse, Jacob Hilton, and John Schulman. Leveraging procedural generation to benchmark reinforcement learning. In *International conference on machine learning*, pages 2048–2056. PMLR, 2020.

[10] Maxime Chevalier-Boisvert, Bolun Dai, Mark Towers, Rodrigo Perez-Vicente, Lucas Willems, Salem Lahlou, Suman Pal, Pablo Samuel Castro, and J K Terry. Minigrid & miniworld: Modular & customizable reinforcement learning environments for goal-oriented tasks. *Advances in Neural Information Processing Systems*, 36:73383–73394, 2023.

[11] Ian Osband, Yotam Doron, Matteo Hessel, John Aslanides, Eren Sezener, Andre Saraiva, Katrina McKinney, Tor Lattimore, Csaba Szepesvari, Satinder Singh, et al. Behaviour suite for reinforcement learning. *arXiv preprint arXiv:1908.03568*, 2019.

[12] Justin Fu, Aviral Kumar, Ofir Nachum, George Tucker, and Sergey Levine. D4RL: Datasets for deep data-driven reinforcement learning. *arXiv preprint arXiv:2004.07219*, 2020.

[13] Vesna Nevistic and James A Primbs. Constrained nonlinear optimal control: a converse hjb approach. *California Institute of Technology, Pasadena, CA*, 91125:96–021, 1996.

[14] Thomas Göhrt, Pavel Osinenko, and Stefan Streif. Converse optimality for discrete-time systems. *IEEE Transactions on Automatic Control*, 65(5):2257–2264, 2019.

[15] Grigory Yamerenko, Sinan Ibrahim, Francisco Moreno-Mora, Pavel Osinenko, and Stefan Streif. Generating informative benchmarks for reinforcement learning. *IEEE Control Systems Letters*, 2025.

[16] Sinan Ibrahim, Grégoire Ouerdane, Hadi Salloum, Henni Ouerdane, Stefan Streif, and Pavel Osinenko. Benchmarking reinforcement learning via stochastic converse optimality: Generating systems with known optimal policies. *arXiv preprint arXiv:2603.17631*, 2026.

[17] Frank H Clarke. *Optimization and nonsmooth analysis*. SIAM, 1990.

[18] Martino Bardi, Italo Capuzzo Dolcetta, et al. *Optimal control and viscosity solutions of Hamilton-Jacobi-Bellman equations*, volume 12. Springer, 1997.

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[19] Michael G Crandall and Pierre-Louis Lions. Viscosity solutions of hamilton-jacobi equations. *Transactions of the American mathematical society*, 277(1):1–42, 1983.

[20] Aleksei Fedorovich Filippov. Differential equations with discontinuous right-hand side. *Matematicheskii sbornik*, 93(1):99–128, 1960.

[21] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[22] Steven E Shreve and Dimitri P Bertsekas. Universally measurable policies in dynamic programming. *Mathematics of Operations Research*, 4(1):15–30, 1979.

[23] Francis Clarke. Nonsmooth analysis in control theory: a survey. *European Journal of Control*, 7(2-3):145–159, 2001.

[24] AF Filippov. Existence and general properties of solutions of discontinuous systems. In *Differential Equations with Discontinuous Righthand Sides*, pages 48–122. Springer, 1988.

[25] Wendell H Fleming and H Mete Soner. *Controlled Markov processes and viscosity solutions*. Springer, 2006.

[26] A. F. Filippov. *Differential Equations with Discontinuous Righthand Sides*. Kluwer, 1988.

[27] Charalambos D Aliprantis and Kim C Border. *Infinite dimensional analysis: a hitchhiker's guide*. Springer, 2006.

A Reproducibility and Released Assets

The framework is intended to be reproducible as a seedable benchmark generator rather than only as a fixed list of hand-authored environments. The public project page: `https://gtfrl.github.io/Benchmarking-of-Reinforcement-Learning-via-Converse-Optimality/`. hosts the paper, code links, fixtures, experimental summaries, and visual diagnostics. A complete release contains generator code, random seeds for sampled fixtures, validation grids, accepted and rejected instance metadata, CRN evaluation schedules, baseline configurations, software versions, and scripts for reproducing the tables.

For the experiments summarized in the main text, the relevant reproducibility objects are the certified fixture parameters, the validation tolerance, the horizon, the start-state distribution, the disturbance streams, and the baseline hyperparameter files. Since the oracle is explicit, independent reproduction can verify both the reported returns and the Bellman consistency of each exported environment.

B Impact Statement

This work aims to improve the reliability and interpretability of RL experimentation. Its main positive impact is methodological: it can help researchers distinguish relative performance from distance to optimality and design more diagnostic experiments. Potential negative impact is limited but not absent. If known-answer fixtures are overused as narrow leaderboards, they could encourage overfitting to generated families or create unwarranted confidence about deployment readiness. We therefore frame the method as a complement to broader benchmarks and real-world validation, not a substitute for them.

C Main Theoretical Results and Proofs

A benchmark fixture is useful only if we can verify that a given pair (V, π) is indeed optimal. The natural verification tool is the *Bellman residual*. For any candidate value function V,

$$\mathcal{R}_V(s,a) \;:=\; c(s,a) + \gamma \int_{\mathcal{S}} V(s')\, P(\mathrm{d}s' \mid s,a) - V(s).$$

The residual measures how much V fails to satisfy the Bellman equation at a given state-action pair. If $V = V^\star$, then $\inf_a \mathcal{R}_V(s,a) = 0$ for all s and the infimum is attained by any optimal action. More

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generally, the residual provides a pointwise certificate that can be checked without solving the full MDP. Proposition 1 makes this precise.

Definition 1. (Bellman operators). For a given MDP $(\mathcal{S}, \mathcal{A}, P, c, \gamma)$, we define the *optimal Bellman operator* T and the *policy evaluation operator* T^π as follows. For any function $V : \mathcal{S} \to \mathbb{R}$,

$$(TV)(s) \coloneqq \inf_{a\in\mathcal{A}} \Big\{ c(s,a) + \gamma \int_{\mathcal{S}} V(s') P(\mathrm{d}s'|s,a) \Big\}, \quad (T^\pi V)(s) \coloneqq c(s,\pi(s)) + \gamma \int_{\mathcal{S}} V(s') P(\mathrm{d}s'|s,\pi(s)).$$

The Bellman operators formalize two fundamental questions in dynamic programming. The *optimal Bellman operator* T asks: "What is the smallest total cost that can be achieved from state s, assuming that the future cost from the next state onward is given by V?". The true optimal value function $V^\star$ is the unique fixed point of this operator, i.e. $V^\star = TV^\star$; this is the Bellman equation. The *policy evaluation operator* T^π asks: "If I follow policy π forever, what total cost will I incur from state s if the future cost is evaluated according to V?". The value function of π is its unique fixed point: $V^\pi = T^\pi V^\pi$.

Proposition 1. (Optimality certificate). Let $\mathbb{B}_w$ be the weighted sup-norm Banach space defined by $\mathbb{B}_w \coloneqq \{V : \mathcal{S} \to \mathbb{R} \text{ measurable } | \|V\|_w < \infty\}$, with $\|V\|_w = \sup_{s\in\mathcal{S}} \frac{|V(s)|}{w(s)}$ and w some suitable weight function. Let T and T^π be the Bellman operators from Definition 1, well-defined on $\mathbb{B}_w$, and admitting unique fixed points $V^\star$ and V^π. For any locally Lipschitz function $V \in \mathbb{B}_w$ and any admissible measurable policy π, the following are equivalent:

1. $V = V^\star = V^\pi$; i.e., V is the true optimal value function and π is an optimal policy.
2. For all $s \in \mathcal{S}$ and all $a \in \mathcal{A}$, we have $\mathcal{R}_V(s,\pi(s)) = 0$ and $\mathcal{R}_V(s,a) \geq 0$.

Proof. We prove the equivalence.

$(1) \Rightarrow (2)$. Assume that $V = V^\star = V^\pi$, and that π is an optimal policy. Since $V^\star$ is the optimal value function, it satisfies the Bellman equation

$$V^\star(s) = \inf_{a\in\mathcal{A}} \left\{ c(s,a) + \gamma \int_{\mathcal{S}} V^\star(s')\, P(\mathrm{d}s' \mid s,a) \right\}.$$

Therefore, for every $(s,a) \in \mathcal{S} \times \mathcal{A}$,

$$c(s,a) + \gamma \int_{\mathcal{S}} V^\star(s')\, P(\mathrm{d}s' \mid s,a) - V^\star(s) \geq 0,$$

that is,

$$\mathcal{R}_V(s,a) \geq 0.$$

Since π is optimal and $V^\pi = V^\star$, the policy π attains the Bellman infimum at every state:

$$V^\star(s) = c(s,\pi(s)) + \gamma \int_{\mathcal{S}} V^\star(s')\, P(\mathrm{d}s' \mid s,\pi(s)).$$

Hence, $\mathcal{R}_V(s,\pi(s)) = 0$. Thus condition (2) holds.

$(2) \Rightarrow (1)$. Assume now that for all $s \in \mathcal{S}$ and all $a \in \mathcal{A}$, $\mathcal{R}_V(s,a) \geq 0$, and $\mathcal{R}_V(s,\pi(s)) = 0$. By definition of the residual, the inequality

$$\mathcal{R}_V(s,a) \geq 0$$

means that

$$V(s) \leq c(s,a) + \gamma \int_{\mathcal{S}} V(s')\, P(\mathrm{d}s' \mid s,a) \qquad \forall a \in \mathcal{A}.$$

Taking the infimum over a gives

$$V(s) \leq (TV)(s).$$

On the other hand, evaluating the residual at the action $\pi(s)$ and using

$$\mathcal{R}_V(s,\pi(s)) = 0,$$

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we obtain

$$V(s) = c(s, \pi(s)) + \gamma \int_{\mathcal{S}} V(s')\, P(\mathrm{d}s' \mid s, \pi(s)) = (T^\pi V)(s).$$

Since

$$(TV)(s) = \inf_{a \in \mathcal{A}} \left\{ c(s,a) + \gamma \int_{\mathcal{S}} V(s')\, P(\mathrm{d}s' \mid s, a) \right\},$$

and the action $a = \pi(s)$ achieves equality, we conclude that

$$(TV)(s) = (T^\pi V)(s) = V(s).$$

Therefore,

$$V = TV \qquad \text{and} \qquad V = T^\pi V.$$

The Bellman operators T and T^π each admit unique fixed points. Hence V must coincide simultaneously with the optimal value function and the value function associated with π:

$$V = V^\star = V^\pi.$$

Therefore π is an optimal policy and condition (1) holds.

This completes the proof. □

Condition 2 of Proposition 1 requires the residual to vanish on the candidate policy and to be non-negative everywhere else. Together they force V to be the fixed point of the optimal Bellman operator and π to attain the minimum. When V is smooth and the action space is unconstrained, the residual inequality reduces to the classical Hamilton–Jacobi–Bellman equation together with a first-order stationarity condition; when V is only locally Lipschitz, the same logic applies with Clarke subdifferentials replacing classical gradients (see Appendix G). With the certificate in hand, we can now define the central object of the benchmarking framework.

C.1 Results for the Stochastic Discrete-Time Setting

To guarantee that the stochastic Bellman equation is well-posed and that the first-order optimality condition is valid, we consider the following set of assumptions [16].

Assumption 1 (Well-posedness and regularity). The following holds on $\mathbb{R}^n$:

1. $q : \mathbb{R}^n \to \mathbb{R}_+$ is continuous.
2. $\rho : \mathbb{R}^m \to \mathbb{R}_+$ is C^2 and μ-strongly convex: $\nabla^2 \rho(a) \succeq \mu I$ for all a.
3. f, g are locally Lipschitz with at most polynomial growth.
4. The candidate value function $V : \mathbb{R}^n \to \mathbb{R}$ is C^2 and convex.

Assumption 2 (Growth and integrability). There exist constants $\eta > 0$ and an integer $\ell \geq 1$ such that for all $s \in \mathbb{R}^n$,

$$|V(s)| + \|\nabla V(s)\| + \|\nabla^2 V(s)\| \leq \eta\big(1 + \|s\|^\ell\big),$$

and, for each fixed $(s, a) \in \mathbb{R}^n \times \mathbb{R}^m$, the expectations $\mathbb{E}[\|\nabla V(f(s) + g(s)a + w)\|]$ and $\mathbb{E}[\|\nabla^2 V(f(s) + g(s)a + w)\|]$ are finite under $w \sim \mathcal{N}(0, \Sigma)$.

Lemma. Under Assumptions 1 and 2, for all $s \in \mathbb{R}^n$ and $a \in \mathbb{R}^m$,

$$\nabla_a \mathbb{E}[V(f(s) + g(s)a + w)] = \mathbb{E}\big[g(s)^\top \nabla V(f(s) + g(s)a + w)\big]\,.$$

Proof. Fix $s \in \mathbb{R}^n$ and define $\phi(a, w) = V(f(s) + g(s)a + w)$. By the chain rule, we have

$$\nabla_a \phi(a, w) = g(s)^\top \nabla V(f(s) + g(s)a + w)$$

To interchange the derivative and the expectation, we verify the conditions of the dominated convergence theorem. Let $e_1, \ldots, e_m$ be the canonical basis vectors of $\mathbb{R}^m$. Without loss of generality, we choose a sequence $\{t_\nu\}$ converging to 0, such that we can define sequences $\{a_\nu^{(i)}\}$ through

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$a_\nu^{(i)} = a + t_\nu e_i$ converging to a along each coordinate axis i. By the mean value theorem, for each axis i, there exists $\xi_\nu^{(i)}$ on the line segment between $a_\nu^{(i)}$ and a such that

$$\phi(a_\nu^{(i)}, w) - \phi(a, w) = \nabla_a \phi(\xi_\nu^{(i)}, w) \cdot (a_\nu^{(i)} - a),$$
$$\text{and } |\phi(a_\nu^{(i)}, w) - \phi(a, w)| \leq \|\nabla_a \phi(\xi_\nu^{(i)}, w)\| \cdot \|a_\nu^{(i)} - a\|.$$

Now, let us bound $\|\nabla_a \phi(\xi_\nu^{(i)}, w)\|$ for each axis i. From Assumption 2, we obtain:

$$\|\nabla_a \phi(\xi_\nu^{(i)}, w)\| \leq \|g(s)\| \cdot \|\nabla V(f(s) + g(s)\xi_\nu^{(i)} + w)\|,$$

and hence,

$$\|\nabla_a \phi(\xi_\nu^{(i)}, w)\| \leq \|g(s)\| \cdot \eta(1 + \|f(s) + g(s)\xi_\nu^{(i)} + w\|^\ell).$$

Since $\xi_\nu^{(i)}$ lies in a bounded neighborhood of a (for ν sufficiently large), and f, g have at most polynomial growth (Assumption 1.3), there exist constants $C_1(s)$ and $C_2(s)$ such that

$$\|f(s) + g(s)\xi_\nu^{(i)} + w\| \leq C_1(s) + C_2(s)\|\xi_\nu^{(i)}\| + \|w\| \leq C(s, a) + \|w\|,$$

where $C(s, a)$ is independent of ν and $\xi_\nu^{(i)}$ (i.e., uniform over the neighborhood of a). This uniformity ensures the subsequent growth bound

$$\|\nabla_a \phi(\xi_\nu^{(i)}, w)\| \leq \eta \|g(s)\|(1 + (C(s, a) + \|w\|)^\ell) \leq K(s, a)(1 + \|w\|^\ell),$$

where $K(s, a)(1 + \|w\|^\ell)$ is independent of ν. This provides a single bound applicable to the entire sequence. The growth bound $\beta : w \mapsto K(s, a)(1 + \|w\|^l)$ is measurable.

Let μ_Σ be the Gaussian measure. By definition of the expectation, the Lebesgue integral of β weighted by the Gaussian density is

$$\int_{\mathbb{R}^n} \beta(w) d\mu_\Sigma(w) = \mathbb{E}[K(s, a)(1 + \|w\|^l)].$$

The linearity of the expectation yields

$$\mathbb{E}[K(s, a)(1 + \|w\|^l)] = K(s, a)(1 + \mathbb{E}[\|w\|^l]),$$

and hence,

$$\int_{\mathbb{R}^n} \beta(w) d\mu_\Sigma(w) = K(s, a) \left(1 + \int_{\mathbb{R}^n} \|w\|^\ell d\mu_\Sigma(w)\right).$$

As all Gaussian moments are finite, we have:

$$\int_{\mathbb{R}^n} \|w\|^\ell d\mu_\Sigma(w) = \mathbb{E}[\|w\|^\ell] < \infty$$

for all $\ell > 0$. Hence,

$$\int_{\mathbb{R}^n} \beta(w) d\mu_\Sigma(w) < \infty,$$

and the growth bound β is Lebesgue-integrable with respect to the Gaussian measure. Since $\xi_\nu^{(i)} \to a$, continuity gives $\nabla_a \phi(\xi_\nu^{(i)}, w) \to \nabla_a \phi(a, w)$ pointwise. Hence, the bound

$$\|\nabla_a \phi(\xi_\nu^{(i)}, w)\| \leq K(s, a)(1 + \|w\|^\ell)$$

provides a suitable dominating function, and by the dominated convergence theorem, we eventually obtain

$$\lim_{\nu \to \infty} \mathbb{E}[\nabla_a \phi(\xi_\nu^{(i)}, w)] = \mathbb{E}[\nabla_a \phi(a, w)]. \quad (*)$$

On the other hand, by definition of the gradient, we have

$$\nabla_a \mathbb{E}[\phi(a, w)] = \Big(\frac{\partial}{\partial a_1} \mathbb{E}[\phi(a, w)], \ldots, \frac{\partial}{\partial a_m} \mathbb{E}[\phi(a, w)]\Big).$$

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The directional derivative along the i-th axis is

$$\frac{\partial}{\partial a_i}\mathbb{E}[\phi(a,w)] = \lim_{\nu\to\infty}\mathbb{E}\Big[\frac{\phi(a+t_\nu e_i,w)-\phi(a,w)}{t_\nu}\Big].$$

Along the i-th axis, we also have

$$\frac{\phi(a+t_\nu e_i,w)-\phi(a,w)}{t_\nu} = \nabla_a\phi(\xi_\nu^{(i)},w)\cdot e_i.$$

Combining these expressions yields

$$\lim_{\nu\to\infty}\mathbb{E}\Big[\frac{\phi(a+t_\nu e_i,w)-\phi(a,w)}{t_\nu}\Big] = \lim_{\nu\to\infty}\mathbb{E}[\nabla_a\phi(\xi_\nu^{(i)},w)\cdot e_i] = \mathbb{E}[\nabla_a\phi(a,w)\cdot e_i],$$

where the last equality follows from the dominated convergence argument $(*)$ above. But, $\nabla_a\phi(a,w)\cdot e_i = \frac{\partial}{\partial a_i}\phi(a,w)$. Therefore, $\frac{\partial}{\partial a_i}\mathbb{E}[\phi(a,w)] = \mathbb{E}[\frac{\partial}{\partial a_i}\phi(a,w)]$. Finally, we get

$$\begin{aligned}\nabla_a\mathbb{E}[\phi(a,w)] &= \Big(\mathbb{E}[\frac{\partial}{\partial a_1}\phi(a,w)],\dots,\mathbb{E}[\frac{\partial}{\partial a_m}\phi(a,w)]\Big) = \\ &= \mathbb{E}\Big(\frac{\partial\phi}{\partial a_1}(a,w),\dots,\frac{\partial\phi}{\partial a_m}(a,w)\Big) = \mathbb{E}[\nabla_a\phi(a,w)].\end{aligned}$$

□

Theorem (Stochastic converse optimality). Fix a stage cost $c(s,a) = q(s)+\rho(a)$ with q,ρ satisfying Assumption 1, discount factor $\gamma\in(0,1)$, noise covariance $\Sigma\succ 0$, input coupling $g:\mathbb{R}^n\to\mathbb{R}^{n\times m}$, and a candidate value function $V^*\in C^2(\mathbb{R}^n)$ satisfying Assumption 2. Then there exists a drift $f:\mathbb{R}^n\to\mathbb{R}^n$ (i.e. a system $s_{k+1} = f(s_k)+g(s_k)a_k+w_k$ with $w_k\sim\mathcal{N}(0,\Sigma)$) for which V^* is the optimal value function if and only if there exists a policy $\pi^*:\mathbb{R}^n\to\mathbb{R}^m$ such that for every $s\in\mathbb{R}^n$:

$$\nabla_a\Big[\rho(a)+\gamma\,\mathbb{E}\big[V^*\big(f(s)+g(s)a+w\big)\big]\Big]\Big|_{a=a^*} = 0, \tag{2}$$

$$V^*(s) = q(s)+\rho(a^*)+\gamma\mathbb{E}\big[V^*\big(f(s)+g(s)a^*+w\big)\big]. \tag{3}$$

When such π^* exists, it is an optimal policy and V^* satisfies the stochastic Bellman equation given in Subsection 5.2 for all $s\in\mathbb{R}^n$.

Proof. Sufficiency. Suppose there exist a policy $\pi^*:\mathbb{R}^n\to\mathbb{R}^m$ and a drift $f:\mathbb{R}^n\to\mathbb{R}^n$ such that for every $s\in\mathbb{R}^n$, equations (2) and (3) hold, and $\pi^*(s_k) = a_k^*$ for all $s_k\in\mathbb{R}^n$. For a fixed $s\in\mathbb{R}^n$ we define the Bellman integrand:

$$\Phi_s(a) := q(s)+\rho(a)+\gamma\mathbb{E}\big[V^*(f(s)+g(s)a+w)\big]$$

By the Lemma, we have:

$$\nabla_a\Phi_s(a) = \nabla_a\rho(a)+\gamma\mathbb{E}\big[g(s)^\top\nabla V^*(f(s)+g(s)a+w)\big].$$

Equations (4)-(5) give $\nabla_a\Phi_s(a)|_{a=a^*} = 0$ and imply that the Bellman identity holds with action a^*. In particular,

$$V^*(s) = q(s)+\rho(a^*)+\gamma\mathbb{E}\big[V^*(f(s)+g(s)a^*+w)\big] = \Phi_s(a^*).$$

Since the Bellman equation involves the minimization of $\Phi_s(a)$ with respect to a, it follows that this minimum is attained at a^*. Hence:

$$\begin{aligned}\min_{a\in\mathbb{R}^m}\Phi_s(a) &= \Phi_s(a^*)\\ &= q(s)+\rho(a^*)+\gamma\mathbb{E}\big[V^*(f(s)+g(s)a^*+w)\big]\\ &= V^*(s).\end{aligned}$$

Thus, π^* is an optimal policy and V^* is the optimal value function for the drift f.

Now fix any stationary policy π and define its Bellman operator:

$$(T^\pi V)(s) := q(s)+\rho(\pi(s))+\gamma\mathbb{E}\big[V\big(f(s)+g(s)\pi(s)+w\big)\big].$$

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By definition of the minimum, $\min_a \Phi_s(a) \le \Phi_s(\pi(s))$, hence $V^*(s) \le (T^\pi V^*)(s)$ for all s. Applying T^π to both sides of $V^* \le T^\pi V^*$ and using the monotonicity of T^π (which follows from the positivity of costs), we obtain $T^\pi V^* \le (T^\pi)^2 V^*$. By induction, $V^* \le (T^\pi)^k V^*$ for all $k \in \mathbb{N}$. Since T^π is a γ-contraction in an appropriate weighted norm (e.g. the sup norm under bounded costs, or a weighted norm under the discounted criterion), then by the Banach fixed-point theorem, $(T^\pi)^k V^* \to V^\pi$, the value function of policy π. Taking limits preserves the inequality, yielding $V^* \le V^\pi$ on $\mathbb{R}^n$, proving the optimality of V^* and π^*.

Necessity. Assume V^* optimal, so for each s the minimizer $\pi^*(s)$ of the Bellman equation exists. By the measurable selection theorem [22], π^* is chosen to be measurable, and hence admissible as a feedback policy. At a minimum, we have $\nabla_a \Phi_s(\pi^*(s)) = 0$. Lemma 1 justifies interchanging the derivative and the expectation, yielding (2). The Bellman identity (3) holds by the definition of $\pi^*(s) = a^*$ as the minimizer of the Bellman equation. □

D Generator Templates and Derivations

D.1 Continuous-Time Converse Derivation

Recall the continuous-time control-affine system $\dot{s} = f(s) + g(s)a$ with running cost $c(s, a)$. For a candidate value function $V \in C^1$, the Hamiltonian is

$$H(s, a) = c(s, a) + \nabla V(s)^\top \big(f(s) + g(s)a\big).$$

The generator [15] prescribes two chosen smooth fields $H : \mathbb{R}^n \to \mathbb{R}^n$ and $G : \mathbb{R}^n \to \mathbb{R}^{n\times m}$ and defines

$$g(s) = G(s), \tag{4}$$

$$f(s) = \frac{1}{4}\, G(s)\, \big(\mathcal{L}_G V(s)\big)^\top - H(s), \tag{5}$$

$$c(s, a) = \mathcal{L}_H V(s) + \|a\|_2^2. \tag{6}$$

Here $\mathcal{L}_G V(s) = \nabla V(s)^\top G(s)$ is a row vector in $\mathbb{R}^{1\times m}$, and $\mathcal{L}_H V(s) = \nabla V(s)^\top H(s)$ is a scalar.

Substituting (4)–(6) into the Hamiltonian,

$$\begin{aligned} H(s, a) &= \mathcal{L}_H V(s) + \|a\|_2^2 + \nabla V(s)^\top \Big(\frac{1}{4}\, G(s) \big(\mathcal{L}_G V(s)\big)^\top - H(s) + G(s)\, a\Big) \\ &= \mathcal{L}_H V(s) + \|a\|_2^2 + \frac{1}{4}\, \nabla V(s)^\top G(s) \big(\mathcal{L}_G V(s)\big)^\top - \nabla V(s)^\top H(s) + \nabla V(s)^\top G(s)\, a. \end{aligned}$$

Because $\mathcal{L}_G V(s) = \nabla V(s)^\top G(s)$, we have

$$\nabla V(s)^\top G(s) \big(\mathcal{L}_G V(s)\big)^\top = \|\mathcal{L}_G V(s)\|_2^2, \qquad \nabla V(s)^\top H(s) = \mathcal{L}_H V(s).$$

Therefore the terms $\mathcal{L}_H V(s)$ cancel, leaving

$$H(s, a) = \|a\|_2^2 + \frac{1}{4}\, \|\mathcal{L}_G V(s)\|_2^2 + \big(\mathcal{L}_G V(s)\big)\, a.$$

The right-hand side is a quadratic form in a. Completing the square,

$$H(s, a) = \big\|\, a + \tfrac{1}{2} \big(\mathcal{L}_G V(s)\big)^\top \big\|_2^2.$$

The unique minimiser of this expression is obtained by making the squared norm vanish, i.e.

$$a^\star(s) = -\tfrac{1}{2}\, \big(\mathcal{L}_G V(s)\big)^\top,$$

which is exactly the equation given in the main text. When V and the constructed cost are positive definite, Proposition 1 guarantees that V is the globally optimal value function for the generated system.

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D.2 Deterministic Discrete-Time Homotopy Construction

Consider deterministic dynamics

$$s_{k+1} = f(s_k) + g(s_k)a_k$$

with separable cost $c(s, a) = q(s) + \rho(a)$ and discount factor absorbed into the notation for this subsection. Göhrt et al. [14] show that a candidate pair $(V^\star, q)$ is admissible if and only if

$$V^\star(s) - q(s) \geq 0 \tag{7}$$

on the state domain. The sufficiency argument is constructive. The Bellman equation requires

$$V^\star(s) - q(s) = \min_a \{\rho(a) + V^\star(f(s) + g(s)a)\}.$$

For each s, the right-hand side is the minimum of a nonnegative function of the next-state point $f(s) + g(s)a$ and the control effort $\rho(a)$. The homotopy construction continuously deforms feasible next-state targets on the level set

$$\{z : V^\star(z) \leq V^\star(s) - q(s)\}$$

while balancing the control cost $\rho(u)$. In the quadratic case,

$$V^\star(s) = s^\top P s, \qquad q(s) = s^\top Q s, \qquad \rho(a) = a^\top R a,$$

with $0 \preceq Q \prec P$ and $R \succ 0$, the admissibility condition is automatic and the construction becomes explicit. Define

$$H(s) = \left(P^{-1} + g(s)R^{-1}g(s)^\top\right)^{-1}.$$

Then any drift of the form

$$f(s) = H(s)^{-1/2} S(s)(P - Q)^{1/2} s, \qquad S(s)^\top S(s) = I, \tag{8}$$

satisfies the deterministic energy identity

$$s^\top (P - Q)s = f(s)^\top H(s) f(s).$$

The optimal feedback is

$$a^\star(s) = -\left(R + g(s)^\top P g(s)\right)^{-1} g(s)^\top P f(s).$$

The orthogonal field $S(s)$ is a design degree of freedom: it changes the drift geometry while preserving the certificate.

D.3 Quadratic-Gaussian Energy Identity Derivation

Let

$$V^\star(s) = s^\top P s + b, \qquad q(s) = s^\top Q s, \qquad \rho(a) = a^\top R a,$$

with $P \succ 0$, $0 \preceq Q \prec P$, and $R \succ 0$. For zero-mean noise with covariance Σ,

$$\mathbb{E}[V^\star(f + ga + w)] = (f + ga)^\top P(f + ga) + \operatorname{tr}(P\Sigma) + b.$$

The Bellman integrand is

$$\begin{aligned}\Phi_s(a) &= s^\top Q s + a^\top R a + \gamma(f + ga)^\top P(f + ga) + \gamma \operatorname{tr}(P\Sigma) + \gamma b \\ &= s^\top Q s + \gamma f^\top P f + 2\gamma a^\top g^\top P f + a^\top (R + \gamma g^\top P g)a + \gamma \operatorname{tr}(P\Sigma) + \gamma b. \end{aligned} \tag{9}$$

Define

$$B_\gamma(s) = R + \gamma g(s)^\top P g(s).$$

Since $B_\gamma(s) \succ 0$, the unique minimizer is

$$a^\star(s) = -\gamma B_\gamma(s)^{-1} g(s)^\top P f(s).$$

Substituting the minimizer gives

$$\begin{aligned}\min_a \Phi_s(a) &= s^\top Q s + \gamma f^\top P f - \gamma^2 f^\top P g B_\gamma^{-1} g^\top P f + \gamma \operatorname{tr}(P\Sigma) + \gamma b \\ &= s^\top Q s + f^\top H_\gamma f + \gamma \operatorname{tr}(P\Sigma) + \gamma b, \end{aligned} \tag{10}$$

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where

$$H_\gamma(s) = \gamma\left(P - \gamma P g(s) B_\gamma(s)^{-1} g(s)^\top P\right).$$

Thus $V^\star(s) = \min_a \Phi_s(a)$ holds if

$$s^\top (P - Q) s = f(s)^\top H_\gamma(s) f(s)$$

and

$$b = \frac{\gamma}{1-\gamma}\operatorname{tr}(P\Sigma).$$

By the matrix inversion lemma,

$$P - \gamma P g (R + \gamma g^\top P g)^{-1} g^\top P = (P^{-1} + \gamma g R^{-1} g^\top)^{-1},$$

so $H_\gamma(s) \succ 0$. This justifies the metric-normalized and square-root parameterizations.

E Non-Affine Raw Actuation and Non-Gaussian Disturbances

This appendix gives a concrete non-affine construction based on a planar aerial-robot interface. The purpose is to show that the ground-truth-first fixture logic is not restricted to raw action-affine interfaces or Gaussian noise.

E.1 Virtual Actuator Coordinates

Let the error state be

$$s = [x, y, z, \phi, \theta, \psi, v_x, v_y, v_z, \omega_x, \omega_y, \omega_z]^\top \in \mathbb{R}^{12}. \tag{11}$$

Let $\bar{u} = [\bar{u}_1, \bar{u}_2, \bar{u}_3, \bar{u}_4]^\top \in \mathbb{R}^4$ denote a virtual thrust-deviation input around hover. With sampling time $\tau > 0$, mass $m > 0$, and inertias $J_x, J_y, J_z > 0$, define

$$g_Q(s) = \tau g_{\mathrm{cont}}(s), \tag{12}$$

where

$$g_{\mathrm{cont}}(s) = \begin{bmatrix} 0_{6\times 4} & & & \\ -\frac{1}{m}\sin\theta & 0 & 0 & 0 \\ \frac{1}{m}\sin\phi\cos\theta & 0 & 0 & 0 \\ \frac{1}{m}(\cos\phi\cos\theta - 1) & 0 & 0 & 0 \\ 0 & \frac{1}{J_x} & 0 & 0 \\ 0 & 0 & \frac{1}{J_y} & 0 \\ 0 & 0 & 0 & \frac{1}{J_z} \end{bmatrix}. \tag{13}$$

The virtual-input dynamics are

$$s_{k+1} = f_p(s_k) + p\, g_Q(s_k)\bar{u}_k + w_k, \qquad p \in (0, 1]. \tag{14}$$

E.2 Raw Action Interface

The RL agent acts through raw command $u \in \mathbb{R}^4$, mapped to virtual input by

$$\bar{u} = \phi(u), \qquad \phi_i(u_i) = u_i + \eta u_i^3, \qquad \eta > 0. \tag{15}$$

Since $\phi_i'(u_i) = 1 + 3\eta u_i^2 > 0$, ϕ is a component-wise C^2 diffeomorphism. The environment seen by RL is

$$s_{k+1} = f_p(s_k) + p\, g_Q(s_k)\phi(u_k) + w_k. \tag{16}$$

For $\eta > 0$, this dynamics is non-affine in the raw action because

$$\frac{\partial^2}{\partial u_i^2}\big(g_Q(s)\phi(u)\big) = 6\eta u_i\, g_{Q,i}(s), \tag{17}$$

where $g_{Q,i}$ is the ith column of g_Q.

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E.3 Cost Pullback and Oracle Preservation

The virtual-action quadratic cost is

$$c_{\bar{u}}(s, \bar{u}) = s^\top Q s + \bar{u}^\top R \bar{u}, \qquad Q \succeq 0,\ R \succ 0. \tag{18}$$

The raw-action cost is the pullback

$$c_u(s, u) = s^\top Q s + \phi(u)^\top R \phi(u). \tag{19}$$

Therefore,

$$\begin{aligned} &\min_u \{c_u(s,u) + \gamma \mathbb{E} V^\star(f_p(s) + p g_Q(s)\phi(u) + w)\} \\ &\quad = \min_{\bar{u}} \left\{ s^\top Q s + \bar{u}^\top R \bar{u} + \gamma \mathbb{E} V^\star(f_p(s) + p g_Q(s)\bar{u} + w) \right\}. \end{aligned} \tag{20}$$

Thus the raw-action MDP is action-coordinate equivalent to the virtual affine MDP. If $\bar{u}_p^\star(s)$ is the certified virtual optimal action, then

$$u_p^\star(s) = \phi^{-1}(\bar{u}_p^\star(s)). \tag{21}$$

For $V^\star(s) = s^\top P s + b$, define

$$B_Q^{(p)}(s) = R + \gamma p^2 g_Q(s)^\top P g_Q(s), \tag{22}$$

$$H_Q^{(p)}(s) = \gamma \Big(P - \gamma p^2 P g_Q(s) (B_Q^{(p)}(s))^{-1} g_Q(s)^\top P \Big). \tag{23}$$

Choose

$$f_p(s) = \big(H_Q^{(p)}(s)\big)^{-1/2} S_p(s) (P - Q)^{1/2} s, \qquad S_p(s)^\top S_p(s) = I. \tag{24}$$

Then

$$f_p(s)^\top H_Q^{(p)}(s) f_p(s) = s^\top (P - Q) s, \tag{25}$$

so the Bellman energy identity holds exactly. The optimal virtual input is

$$\bar{u}_p^\star(s) = -\gamma p \big(B_Q^{(p)}(s)\big)^{-1} g_Q(s)^\top P f_p(s). \tag{26}$$

E.4 Finite-Covariance Non-Gaussian Gusts

Let w_k be any zero-mean disturbance with finite covariance Σ. For example,

$$w_k \sim (1-\epsilon)\mathcal{N}(0, \Sigma_1) + \epsilon \mathcal{N}(0, \Sigma_2), \qquad \Sigma = (1-\epsilon)\Sigma_1 + \epsilon \Sigma_2. \tag{27}$$

The Gaussian mixture is only an illustration; the certificate does not require Gaussianity. Since $V^\star$ is quadratic,

$$\mathbb{E}[V^\star(z + w)] = z^\top P z + \operatorname{tr}(P\Sigma) + b. \tag{28}$$

The Bellman minimizer therefore depends on the disturbance only through its covariance. The value offset remains

$$b = \frac{\gamma}{1-\gamma} \operatorname{tr}(P\Sigma). \tag{29}$$

F Benchmark Generation, Experimental Implementation, and Reproducibility

This appendix describes the validation, implementation, and reproducibility protocol used for the reported benchmark experiments. The released benchmark code is organized into two self-contained experimental modes. Mode I implements continuous-time sensitivity studies and symbolic generator demonstrations, while Mode II implements native discrete-time oracle-referenced fixtures. Both modes follow the same separation principle: the learner interacts only with the exposed environment, whereas the certified value function, optimal policy, validation traces, and oracle diagnostics are used only for benchmark validation and post-training evaluation.

Mode	Implementation path
Mode I: continuous-time sensitivity and generator studies	`codemode1/`
Mode II: native discrete-time oracle-referenced fixtures	`Codemode2/Code/`

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F.1 Generator Validation and Experimental Implementation

Validation checks for native discrete-time fixtures. The stochastic discrete-time generator in Algorithm 1 produces each fixture together with a validation trace. Each exported instance undergoes three checks.

1. **Symmetric positive-definiteness of the Bellman action metric.** For the quadratic–Gaussian construction, the Bellman integrand contains

$$B_\gamma(s) = R + \gamma g(s)^\top P g(s).$$

The condition $B_\gamma(s) \succ 0$ is required for two reasons. First, it makes the Bellman integrand strictly convex in the action, so the first-order optimality condition identifies the unique global minimizer $a^\star(s)$. Second, it keeps the discounted metric

$$H_\gamma(s) = \gamma \left(P - \gamma P g(s) B_\gamma(s)^{-1} g(s)^\top P\right)$$

positive definite, which is necessary for the energy equation

$$s^\top (P - Q) s = f(s)^\top H_\gamma(s) f(s)$$

to be well-posed and numerically stable. In the implementation, this condition is checked on a prescribed bounded validation domain $\mathcal{D}$. Instances that violate the numerical lower-eigenvalue tolerance are rejected.

2. **Bellman residual.** The Bellman residual check verifies that the constructed dynamics, costs, value, and policy satisfy the certified optimality identity. For each grid point $s \in \mathcal{G}$, the implementation computes

$$\delta_{\mathrm{Bell}}(s) = \left| V^\star(s) - \left(q(s) + \rho(a^\star(s)) + \gamma \mathbb{E}[V^\star(f(s) + g(s)a^\star(s) + w)]\right)\right|.$$

For the quadratic–Gaussian case, the expectation is evaluated analytically through the covariance term. An instance passes this check if

$$\max_{s \in \mathcal{G}} \delta_{\mathrm{Bell}}(s) < \varepsilon_{\mathrm{Bell}},$$

where $\varepsilon_{\mathrm{Bell}}$ is the numerical tolerance recorded with the fixture. This check prevents implementation-level errors in the oracle, drift, cost, or noise offset.

3. **Closed-loop boundedness.** The closed-loop system under $\pi^\star$ is simulated over the validation horizon and domain. This verifies that the constructed instance is not only algebraically certified but also numerically usable for finite-horizon RL evaluation. Trajectories that diverge, leave the accepted validation domain, or accumulate numerical instability cause the candidate fixture to be rejected.

Validated native discrete-time fixtures are specified through version-controlled configuration artifacts containing the generator parameters, cost matrices, disturbance covariance, discount factor, oracle definitions for $V^\star$ and $\pi^\star$, validation tolerances, and evaluation seeds.

Algorithm 1 Converse-Optimal Benchmark Generator (quadratic–Gaussian case)

Input: family type, parameter ranges, number of instances N, parameterization (`sqrt` or `metric`), nonlinearity knob α, seeds
for each hyperparameter tuple ψ **do**
  Sample parameter vector ψ_i from the family-specific distribution P_Ψ
  Extract $(n, m, Q, R, P, \gamma, \Sigma)$ and input map g from ψ_i
  Verify $B_\gamma(s) = R + \gamma g(s)^\top P g(s) \succ 0$ on the validation domain $\mathcal{D}$
  Choose a bounded direction field $\tilde{f}(s)$ or orthogonal field $S(s)$
  Construct f by the selected converse parameterization and compute $a^\star(s)$
  Validate the Bellman identity on $\mathcal{G}$ and test closed-loop boundedness under $a^\star$
  Export the accepted fixture with oracle accessors, validation metadata, and evaluation seeds
end for
Output: accepted fixture set, validator, oracle evaluators, baseline scripts, and plotting scripts

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Mode I: continuous-time sensitivity experiment. Mode I evaluates whether an RL training pipeline remains stable under controlled variation of certified continuous-time systems. The main sensitivity experiment compares two PPO pipelines that differ only in the PyTorch optimizer used inside Stable-Baselines3, namely Adam and RAdam. The system-generation schedule is fixed by generator seeds, while neural-network initialization and optimizer stochasticity are not frozen. Thus, the experiment measures optimizer-induced outcome variability under the same generated-system schedule rather than a single deterministic training trace.

Each generated Mode-I sensitivity instance is a three-dimensional, three-input continuous-time system

$$\dot{s} = f_\theta(s) + g_\theta(s)a, \qquad s \in \mathbb{R}^3, \quad a \in \mathbb{R}^3.$$

The generator uses the converse-HJB template

$$f_\theta(s) = -\xi_\theta(s) + \frac{1}{4} g_\theta(s) g_\theta(s)^\top \nabla V(s), \qquad V(s) = \|s\|_2^2, \qquad \xi_\theta(s) = 1.2\,\nabla V(s).$$

The input matrix $g_\theta(s)$ is generated symbolically using a truncated trigonometric expansion. The continuous-time system is exposed to the RL agent through a Gymnasium wrapper using explicit Euler integration. The reported optimizer-sensitivity study uses horizon 200, integration step 0.01, initial state $s_0 = (-1.5, -1.5, -1.5)^\top$, action bounds $[-10, 10]^3$, observation bounds $[-10, 10]^3$, and training reward

$$r(s, a) = -\|s\|_2^2 - \|a\|_2^2,$$

with numerical clipping to keep training traces bounded.

The Adam–RAdam comparison contains 80 generated-system runs per optimizer and 160 runs in total. The run schedule is generated from the seed/count pairs

$$(2, 10), \qquad (4, 10), \qquad (5, 20), \qquad (5, 40),$$

where (seed, M) denotes M sequentially generated systems after resetting the generator to the specified seed. The reproduction command is

```
cd "code mode 1"
python3 main.py
```

The TensorBoard directory is `multi_opt_full_experiment/`. Stored CSV outputs for the 160-run study are provided under `Combined_CSVs/`, and the analysis scripts compute the reported variance, histogram, boxplot, and kernel-density diagnostics:

```
python3 analysis_scripts/tensorboard_fscore.py
python3 analysis_scripts/tensorboard_hist.py
python3 analysis_scripts/tensorboard_boxplot.py
python3 analysis_scripts/Final_Rewards_Distribution_KDE.py
```

All PPO hyperparameters are shared between Adam and RAdam:

Quantity	Value	Comment
Policy class	`MlpPolicy`	Stable-Baselines3 PPO
Feature extractor	`CustomMLP`	feature dimension 128
Policy network	$[64, 64]$	shared architecture
Value network	$[64, 64]$	shared architecture
Optimizer	Adam or RAdam	only varied factor
Learning rate	3×10^{-4}	shared
Rollout length	2048	`n_steps`
Batch size	64	shared
Epochs per update	10	shared
Discount	0.99	shared
GAE parameter	0.95	shared
PPO clip range	0.1	shared
Entropy coefficient	0.01	shared
Gradient clipping	0.3	`max_grad_norm`
Training budget	600000 steps	per generated system

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Mode I also includes a symbolic N-crank generator demonstration. This experiment is not used as the main oracle-referenced comparison in the paper; it demonstrates that the continuous-time converse template can generate physically interpretable nonlinear control families. The implemented example uses $N = 4$ coupled cranks with state

$$s = (x_1, y_1, \ldots, x_N, y_N)^\top \in \mathbb{R}^{2N},$$

where x_i and y_i are the angular position and angular velocity of crank i, and u_i is the torque applied at the i-th pivot. Adjacent cranks are coupled through spring-like trigonometric terms. The symbolic construction uses

$$f(s) = \frac{1}{4} G(s) L_G V(s)^\top - H(s), \qquad c(s, u) = L_H V(s) + u^\top u,$$

with analytic optimal policy

$$u^\star(s) = -\frac{1}{2} L_G V(s)^\top.$$

The training interface exposes PPO, A2C, and SAC:

```
python3 train_all_crank.py --run_both
```

and the evaluator compares learned controllers with the analytic optimal controller and the uncontrolled baseline $u = 0$:

```
python3 eval_all_with_sim_cranks.py
```

The evaluator exports state trajectories, control signals, value-function traces, stage-cost traces, animations, and CSV files under `plots/` and `plots/data/`.

Mode II: native discrete-time oracle-referenced fixtures. Mode II uses native discrete-time fixtures, so the environment transition law, oracle, learner interface, and evaluation metrics are defined in the same formalism. This removes the additional ambiguity introduced by choosing a numerical integration step for an underlying continuous-time system.

The Mode-II code contains two reported benchmark families. **ConverseArm** is a stochastic serial planar arm with state-dependent actuation. For n links,

$$s = (\theta_1, \omega_1, \ldots, \theta_n, \omega_n)^\top \in \mathbb{R}^{2n}, \qquad a = (\tau_1, \ldots, \tau_n)^\top \in \mathbb{R}^n.$$

The cost and value oracle are

$$c(s, a) = s^\top Q s + a^\top R a, \qquad V^\star(s) = s^\top P s + c_\Sigma, \qquad c_\Sigma = \frac{\gamma}{1 - \gamma} \operatorname{tr}(P\Sigma).$$

The input matrix is configuration dependent:

$$[g(s)]_{2i,i} = \cos(\theta_i), \qquad i = 1, \ldots, n,$$

with all other entries zero. The transition has the form

$$s_{k+1} = f_p(s_k) + p\, g(s_k) a_k + w_k, \qquad w_k \sim \mathcal{N}(0, \Sigma),$$

where p is the control-authority parameter and f_p is generated by the stochastic quadratic converse construction. The environment returns reward $r_k = -c(s_k, a_k)$, while all oracle diagnostics are computed in the cost convention and converted to reward convention only for plotting when explicitly stated.

NVDEx is the dynamically extended nonholonomic-vehicle family. The implementation name is `NUDEx-v0`; the manuscript uses the family name NVDEx. Each module has state

$$s_i = (x_i, y_i, \phi_i, v_i^{(0)}, \ldots, v_i^{(r_v - 1)}, \omega_i^{(0)}, \ldots, \omega_i^{(r_\omega - 1)})^\top.$$

The kinematic variables evolve as

$$x_{i,k+1} = x_{i,k} + \tau v_{i,k}^{(0)} \cos \phi_{i,k}, \qquad y_{i,k+1} = y_{i,k} + \tau v_{i,k}^{(0)} \sin \phi_{i,k},$$

$$\phi_{i,k+1} = \phi_{i,k} + \tau \omega_{i,k}^{(0)}.$$

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The velocity-extension chains shift forward in discrete time, and the action enters only at the highest derivative. This creates delayed actuation and longer credit-assignment paths while retaining the certified quadratic oracle.

Both Mode-II families expose exact oracle methods through the benchmark object:

$$\texttt{oracle.action}(s) = \pi^\star(s), \qquad \texttt{oracle.value}(s) = V^\star(s), \qquad \texttt{oracle.q_value}(s,a) = Q^\star(s,a).$$

The oracle-referenced baselines are

$$\{\mathrm{PPO}, \mathrm{A2C}, \mathrm{SAC}, \mathrm{TD3}, \mathrm{DDPG}\}.$$

The released scripts use Stable-Baselines3 implementations with the following principal hyperparameters:

Algorithm	Main hyperparameters	Type
PPO	`n_steps`=1024, `batch_size`=256, `n_epochs`=10, lr $= 3\cdot10^{-4}$, `ent_coef`=10^{-4}, `clip`=0.2	On-policy
A2C	`n_steps`=20, lr $= 3\cdot10^{-4}$, `ent_coef`=10^{-4}	On-policy
SAC	`batch_size`=256, `buffer`=700000, `learning_starts`=10000, lr $= 3\cdot10^{-4}$, $\tau = 0.005$, `ent_coef=auto`	Off-policy
TD3	`batch_size`=256, `buffer`=700000, `learning_starts`=25000, lr $= 3\cdot10^{-4}$, $\tau = 0.005$, `policy_delay`=2	Off-policy
DDPG	`batch_size`=256, `buffer`=700000, `learning_starts`=25000, lr $= 3\cdot10^{-4}$, $\tau = 0.005$	Off-policy

The exact fixture grids, algorithm schedules, training budgets, seeds, and difficulty presets used by the released experiments are stored in the YAML configuration files and shell launchers. This avoids hard-coding a separate table in the manuscript that could drift from the executable benchmark configuration.

F.2 Oracle Evaluation, Artifacts, and Reproducibility Checklist

Cost and reward conventions. All oracle definitions are naturally expressed in the cost convention. Since the Gymnasium environments return rewards $r = -c$, the implementation records the convention used by each diagnostic. For a learned deterministic policy $\hat{\pi}$, rollout horizon T, and initial state s_0, the tail-corrected cost gap is

$$\mathrm{Gap}_T^c(\hat{\pi}; s_0) = \frac{\mathbb{E}\left[\sum_{k=0}^{T-1} \gamma^k c(s_k, \hat{\pi}(s_k)) + \gamma^T V^\star(s_T)\right] - V^\star(s_0)}{|V^\star(s_0)| + \epsilon}.$$

This quantity is zero for the certified optimal policy up to numerical precision and is nonnegative for policies worse than the oracle. Equivalently, if $G^\star = -V^\star$ denotes the reward-value oracle, the reward-convention signed gap is

$$\mathrm{Gap}_T^r(\hat{\pi}; s_0) = \frac{\mathbb{E}\left[\sum_{k=0}^{T-1} \gamma^k r(s_k, \hat{\pi}(s_k)) + \gamma^T G^\star(s_T)\right] - G^\star(s_0)}{|G^\star(s_0)| + \epsilon}.$$

In this convention, zero is optimal and more negative values indicate larger suboptimality. The paper reports explicitly whether a plot uses Gap_T^c or Gap_T^r.

The discounted cost regret is

$$\mathrm{Regret}_T^c(\hat{\pi}; s_0) = \mathbb{E}\left[\sum_{k=0}^{T-1} \gamma^k \Big(c(s_k, \hat{\pi}(s_k)) - c(s_k, \pi^\star(s_k))\Big)\right].$$

Lower cost regret is better. If a reward-convention regret is plotted, it is defined by

$$\mathrm{Regret}_T^r = -\mathrm{Regret}_T^c,$$

and the sign convention is stated in the corresponding caption.

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Common-random-number evaluation schedule. For each stochastic fixture θ, the evaluation schedule is generated before policy evaluation:

$$\mathcal{E}_\theta = \left\{ \left(s_0^{(j)}, w_{0:T-1}^{(j)} \right) \right\}_{j=1}^{M}.$$

All algorithms are evaluated on exactly the same $\mathcal{E}_\theta$. If the environment is deterministic, the schedule reduces to matched start states. If the environment is stochastic, the same disturbance stream is replayed for each policy. This gives a paired design: the difference between two algorithms on trajectory j is evaluated under the same initial condition and randomness. In the implementation, evaluation seeds are deterministic offsets of the training seed, for example

$$\text{evaluation seed} = \text{training seed} + 10000, \qquad \text{CRN metric seed} = \text{training seed} + 20000.$$

For NVDEx, the evaluation callback uses deterministic policy actions, evaluates at fixed environment-step intervals, and averages over multiple CRN-matched evaluation episodes.

Confidence intervals are computed by paired bootstrap:

1. Sample trajectory indices $j_1, \ldots, j_M$ with replacement.
2. Recompute the metric difference using the resampled paired trajectories.
3. Repeat for B_{boot} bootstrap replicates.
4. Report the empirical 2.5% and 97.5% quantiles.

The released evaluation schedule includes the seeds and serialized start-state/noise arrays needed to reproduce paired evaluation.

Bellman-consistency diagnostics. The implementation also logs Bellman-consistency quantities using the known value. In the cost convention, a one-step diagnostic is

$$\delta_k^c = V^\star(s_k) - \left(c(s_k, a_k) + \gamma V^\star(s_{k+1}) \right).$$

For oracle actions under the exact transition law, this residual is zero in expectation up to numerical and Monte Carlo error. These diagnostics are used for validation and sanity checking; they are not exposed to the learner during training.

Output artifacts. The released experiments produce artifacts at three levels. At the generator level, they include configuration files, generator seeds, accepted/rejected instance logs, and validation outputs. At the learning level, they include baseline training scripts, hyperparameter files, trained-model directories, TensorBoard traces, and rollout CSVs. At the evaluation level, they include oracle metrics, CRN schedules, bootstrap summaries, heatmaps, trajectory plots, and diagnostic figures.

The main output locations are:

- Mode-I optimizer-sensitivity traces under `multi_opt_full_experiment/` and aggregated CSVs under `Combined_CSVs/`.
- Mode-I statistical plots under `analysis_scripts/`, including variance-test, histogram, boxplot, and KDE diagnostics.
- N-crank trajectories, control traces, value traces, stage-cost traces, animations, and CSVs under `plots/` and `plots/data/`.
- ConverseArm logs, models, rollout CSVs, and figures under the released arm experiment directories.
- NVDEx logs, CSV summaries, and plotting scripts under `nvdexruns/` and its evaluation subdirectories.

Reproducibility checklist. The anonymous project website and repository include:

1. Generator source code and version hash.
2. Fixture/configuration files containing $(P, Q, R, \Sigma, g, f, \gamma, V^\star, \pi^\star)$ or executable equivalents.
3. Generator seeds and accepted/rejected instance logs.
4. Validation-domain specifications and tolerance values.

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5. Common-random-number evaluation schedules.
6. Baseline training scripts and hyperparameter files.
7. Metric scripts for optimality gap, regret, policy error, Bellman residual, and bootstrap intervals.
8. Figure-generation scripts for tables, heatmaps, trajectories, and distributional plots.
9. Environment, Python, CUDA, and dependency versions.
10. Hardware and wall-clock summaries for the reported runs: Intel® Core™ i9-9900 CPU, RTX 3080 12 GB, and 16 GB DDR4 RAM.

This structure makes the benchmark independently auditable. Another group can regenerate or reload fixtures, verify Bellman consistency, train policies without oracle access, and reproduce oracle-referenced evaluation using the same paired start-state and disturbance schedules.

F.3 Additional Numerical Results for the Discrete-Time Stochastic Case

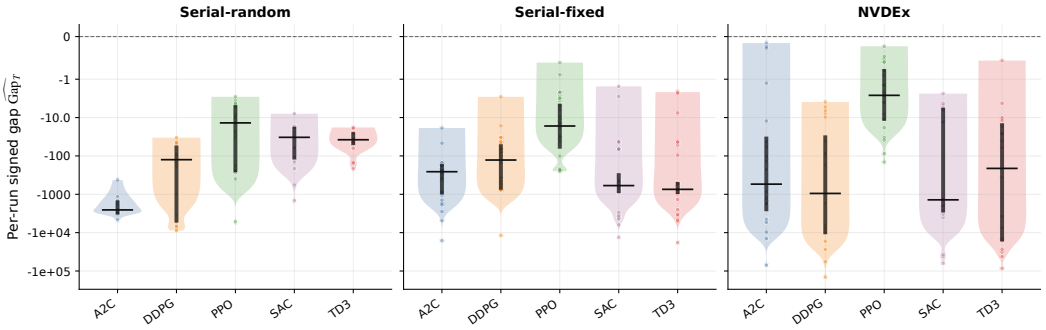


Figure 6: Distribution of reward-convention signed gaps by regime and algorithm. Points correspond to available run/fixture evaluation rows; violin widths show the empirical distribution of signed gaps. The dashed line marks the certified optimum at zero. Black horizontal markers indicate central tendency. Unlike Figure 5, this is not a confidence-interval plot; it is a distributional diagnostic that exposes variability and lower-tail failures.

Table 3: Oracle-referenced comparison on native discrete-time fixtures. Metrics are mean and 95% CI via paired bootstrap under CRN. For signed gap, zero is optimal and values closer to zero are better; for regret, smaller is better.

Experiment	Algorithm	Signed gap	Signed gap 95% CI	Regret	Regret 95% CI
Serial-random	A2C	-2.36e+03	[-2.79e+03, -1.93e+03]	8.24e+03	[6.83e+03, 9.58e+03]
Serial-random	DDPG	-2.29e+03	[-3.83e+03, -9.12e+02]	9.28e+03	[3.50e+03, 1.63e+04]
Serial-random	PPO	-3.32e+02	[-8.45e+02, -4.92e+01]	1.51e+03	[1.92e+02, 3.93e+03]
Serial-random	SAC	-1.65e+02	[-3.47e+02, -4.63e+01]	6.90e+02	[1.72e+02, 1.53e+03]
Serial-random	TD3	**-5.85e+01**	**[-8.64e+01, -3.57e+01]**	**2.14e+02**	**[1.25e+02, 3.35e+02]**
Serial-fixed	A2C	-1.11e+03	[-2.15e+03, -4.58e+02]	5.78e+04	[3.12e+04, 9.23e+04]
Serial-fixed	DDPG	-6.48e+02	[-1.34e+03, -2.51e+02]	3.06e+04	[1.61e+04, 5.22e+04]
Serial-fixed	PPO	**-6.46e+01**	**[-9.56e+01, -3.68e+01]**	**4.08e+03**	**[2.30e+03, 6.14e+03]**
Serial-fixed	SAC	-1.51e+03	[-2.44e+03, -8.00e+02]	9.03e+04	[5.18e+04, 1.33e+05]
Serial-fixed	TD3	-1.53e+03	[-2.69e+03, -7.61e+02]	8.31e+04	[5.05e+04, 1.23e+05]
NVDEx	A2C	-5.60e+03	[-1.33e+04, -1.06e+03]	5.94e+02	[1.34e+02, 1.33e+03]
NVDEx	DDPG	-1.46e+04	[-3.13e+04, -3.37e+03]	3.22e+03	[4.76e+02, 7.39e+03]
NVDEx	PPO	**-1.58e+01**	**[-3.07e+01, -4.57e+00]**	**2.97e+00**	**[7.01e-01, 5.82e+00]**
NVDEx	SAC	-7.76e+03	[-1.56e+04, -1.15e+03]	1.56e+03	[1.13e+02, 3.13e+03]
NVDEx	TD3	-1.22e+04	[-2.24e+04, -4.20e+03]	2.78e+03	[6.96e+02, 5.67e+03]

Figure F.3 shows that the mean values in Table 3 do not fully describe the failure modes. Several algorithm–regime pairs have heavy negative tails, meaning that some evaluated runs or fixtures are not merely slightly suboptimal but orders of magnitude farther from the certified optimum.

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G Non-smooth Value Functions

The smooth constructions in the main text are not the boundary of the framework. This appendix proposes a non-smooth framework for deterministic continuous-time systems. It is useful for generating fixtures with switching surfaces, saturated policies, or discontinuous optimal feedback.

G.1 Background

Overview

Standard RL benchmarks often exhibit non-smooth value functions, making learning challenging. Without ground truth, one cannot distinguish genuine non-smoothness from overfitting. Converse optimality can generate benchmarks with known non-smooth optimal solutions. Previous converse theorems assumed C^2 smoothness, excluding bang-bang control and switching surfaces. We remove this assumption.

When optimal value functions are non-smooth

If the control enters linearly and the cost does not penalize control quadratically, the optimal policy is often bang-bang, and the value function has kinks along switching manifolds where the derivative is not defined [23]. For non-differentiable V^*, the classical HJB equation is ill-defined. Viscosity solutions replace pointwise derivatives with inequalities involving smooth test functions [18, 19]. Consider the HJB equation

$$F(s, V(s), DV(s)) = 0, \quad s \in \Omega.$$

V is a *viscosity subsolution* if for every C^1 test function ϕ and every local maximum s_0 of $V - \phi$, $F(s_0, V(s_0), D\phi(s_0)) \leq 0$. It is a *viscosity supersolution* if for every local minimum s_0 of $V - \phi$, $F(s_0, V(s_0), D\phi(s_0)) \geq 0$. V is a *viscosity solution* if it is both. For optimal control, V^* is the unique viscosity solution of the HJB equation

$$\sup_{a \in \mathbb{A}} \big\{ -\nabla V^*(s) \cdot f(s, a) - c(s, a) \big\} = 0.$$

Clarke subdifferential and non-smooth analysis

For locally Lipschitz V, the Clarke generalized directional derivative is

$$V^\circ(s; v) := \limsup_{y \to s,\, t \downarrow 0} \frac{V(y + tv) - V(y)}{t},$$

and the Clarke subdifferential is

$$\partial_C V(s) := \{\zeta \in \mathbb{R}^n : V^\circ(s; v) \geq \langle \zeta, v \rangle \ \forall v\}.$$

$\partial_C V(s)$ is nonempty, compact, convex; it reduces to $\{\nabla V(s)\}$ where V is smooth, and to the convex hull of limiting gradients at switching surfaces. A locally Lipschitz V is a viscosity solution of the HJB equation iff for every s and every $p \in \partial_C V(s)$ the HJB inequality holds [23]. When the optimal policy is discontinuous, the closed-loop dynamics $\dot{s} = f(s) + g(s)a^*(s)$ is interpreted in the sense of Filippov (differential inclusion) [24].

G.2 Problem Setup and Assumptions

Let the state space be $\mathbb{R}^n$ and the control space be $\mathbb{A} \subset \mathbb{R}^m$. We consider control-affine dynamical systems of the form

$$\dot{s}(t) = f(s(t)) + g(s(t))a(t), \quad s(0) = s_0,$$

where $a(\cdot)$ is an admissible control signal taking values in $\mathbb{A}$.

The performance of a control policy π is evaluated using the infinite-horizon cost functional

$$V^\pi(s_0, u(\cdot)) = \int_0^\infty c(s(t), \pi(s(t))\, dt,$$

where $c : \mathbb{R}^n \times \mathbb{A} \to \mathbb{R}$ is the running cost.

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The optimal value function is defined as

$$V^*(s_0) = \inf_{\pi} V^{\pi}(s_0, \pi(s(t))).$$

In the converse optimality setting, we instead begin with a candidate function $V : \mathbb{R}^n \to \mathbb{R}$ and aim to identify (f, g) such that $V = V^*$.

Assumption 1 (System and control structure).

(a) The control set $\mathbb{A} \subset \mathbb{R}^m$ is compact and nonempty.

(b) The dynamics $f : \mathbb{R}^n \to \mathbb{R}^n$ and the control matrix field $g : \mathbb{R}^n \to \mathbb{R}^{n\times m}$ are continuous and locally Lipschitz-continuous in the state variable, i.e., for any compact set $S \subset \mathbb{R}^n$, there exists $L_{f,S}, L_{g,S} > 0$ such that for all $s_1, s_2 \in S$,

$$\|f(s_1) - f(s_2)\| \le L_{f,S}\|s_1 - s_2\|,$$
$$\|g(s_1) - g(s_2)\| \le L_{g,S}\|s_1 - s_2\|.$$

(c) The running cost $c(s, a)$ is continuous and bounded from below.

(d) The control cost is assumed to be coercive.

Compactness of the control set guarantees the existence of minimizing controls in pointwise optimization problems arising in the HJB equation. Lipschitz continuity of the dynamics ensures existence and uniqueness of trajectories for admissible controls, making the control problem well posed. Boundedness from below of the cost functional prevents pathological cases where the value function diverges to $-\infty$. These assumptions are standard in dynamic programming and HJB theory and are required for the validity of the dynamic programming principle.

Assumption 2 (Value function regularity). The candidate value function $V : \mathbb{R}^n \to \mathbb{R}$ satisfies:

(a) V is continuous and locally Lipschitz.

(b) V is thus differentiable almost everywhere (Rademacher). At points of nondifferentiability, the Clarke subdifferential $\partial_C V(s)$ exists and is nonempty.

(c) For every initial condition $s_0 \in \mathbb{R}^n$, the value function

$$V^*(s_0) := \inf_{a(\cdot)} \int_0^{\infty} c(s(t), a(t))\, dt$$

is well-defined and finite, i.e., $V^*(s_0) < +\infty$.

(d) For the candidate optimal feedback $a^*(s)$ constructed in this work, the corresponding closed-loop trajectory $s(\cdot)$ satisfies

$$\int_0^{\infty} c(s(t), a^*(s(t)))\, dt < +\infty.$$

In general optimal control problems, the value function is rarely smooth due to control constraints, switching surfaces, or state constraints. Local Lipschitz continuity is the natural regularity class for value functions and guarantees almost-everywhere differentiability. The Clarke subdifferential provides a generalized gradient suitable for non-smooth analysis and allows to formulate generalized HJB conditions that are consistent with viscosity solution theory. When V is not differentiable, the classical HJB equation involving ∇V is no longer meaningful. The gradient is replaced by elements of the Clarke subdifferential and corresponds to the viscosity solution formulation of the HJB equation.

Definition 1 (Viscosity HJB condition). The triple (f, g, c) is said to realize V as an optimal value function if the following conditions hold:

(i) **(Subsolution)** For every $s \in \mathbb{R}^n$ there exists $p \in \partial_C V(s)$ such that

$$\inf_{a\in\mathbb{A}} \left\{c(s, a) + p \cdot (f(s) + g(s)a)\right\} \le 0.$$

(ii) **(Supersolution)** For every $s \in \mathbb{R}^n$ and every $p \in \partial_C V(s)$,

$$\inf_{a\in\mathbb{A}} \left\{c(s, a) + p \cdot (f(s) + g(s)a)\right\} \ge 0.$$

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(iii) **(Measurable optimal feedback)** There exists a measurable function $a^* : \mathbb{R}^n \to \mathbb{A}$ and a measurable selection $p(s) \in \partial_C V(s)$ such that for almost every s

$$a^*(s) \in \arg\min_{a \in \mathbb{A}} \big\{ c(s,a) + p(s) \cdot (f(s) + g(s)a) \big\},$$

and the minimum value equals zero for almost every s. The closed-loop system $\dot{s} = f(s) + g(s)a^*(s)$ admits solutions (in the sense of Filippov when a^* is discontinuous).

If the viscosity HJB condition of Definition 1 holds, then V is a viscosity solution of the Hamilton–Jacobi–Bellman equation associated with the control problem (see, e.g., [18, 25]) and, by the uniqueness of viscosity solutions, V is the optimal value function. The feedback a^* given by condition (iii) is thus an optimal policy.

Given a non-smooth candidate V satisfying Assumption 2, three questions naturally arise:

1. **Existence:** Does there exist (f, g, c) meeting the viscosity condition?
2. **Construction:** Can explicit formulas for f, g, c be given in terms of V and its Clarke subdifferential?
3. **Parameterization:** Can the construction be parameterized to produce families of systems that all share the same V?

When V is C^1, $\partial_C V(s) = \{\nabla V(s)\}$ and the viscosity condition collapses to the classical pointwise HJB equation. Hence the non-smooth formulation is a strict generalization of the smooth converse theory studied in earlier work [13, 14].

G.3 Theoretical Results

We now establish the theoretical foundations of non-smooth converse optimality for deterministic control-affine systems. Throughout this section, we assume that the candidate value function V satisfies Assumption 2 (locally Lipschitz, with Clarke subdifferential $\partial_C V(s)$ defined everywhere), and that the system components (f, g, c) satisfy Assumption 1.

Lemma 1. (First-order optimality condition). Let Assumptions 1–2 hold and let the stage cost be $c(s,a) = q(s) + \rho(a)$ with $\rho \in C^2$ strictly convex. Assume that the action space is the whole space $\mathbb{A} = \mathbb{R}^m$ (no control constraints). Assume that V is a viscosity solution of the HJB equation and that there exist measurable functions $a^* : \mathbb{R}^n \to \mathbb{R}^m$ and $p : \mathbb{R}^n \to \mathbb{R}^n$, $p(s) \in \partial_C V(s)$, such that the minimum in the viscosity HJB condition is attained and equals zero for almost every s. Then, for almost every s, the following first-order condition holds:

$$\nabla_a \rho(a)\big|_{a=a^*(s)} + g(s)^\top p(s) = 0. \tag{30}$$

Proof. Fix any $s \in \mathbb{R}^n$ for which the minimum in the viscosity HJB condition is attained and equals zero, and let $p = p(s) \in \partial_C V(s)$ be the corresponding generalized gradient. Define the function

$$\Phi_s(a) \coloneqq q(s) + \rho(a) + p \cdot \big(f(s) + g(s)a\big).$$

By assumption, $a^*(s)$ is a global minimizer of Φ_s over $\mathbb{R}^m$ and $\Phi_s(a^*(s)) = 0$. Since $q(s)$ and $p \cdot f(s)$ do not depend on a, minimizing Φ_s is equivalent to minimizing

$$h(a) \coloneqq \rho(a) + p \cdot g(s)a, \qquad a \in \mathbb{R}^m.$$

The function ρ is strictly convex and C^2 by Assumption 1. Moreover, ρ is coercive, so $h(a) \to \infty$ as $\|a\| \to \infty$. Hence the sublevel sets of h are compact, and a global minimum of h exists. Strict convexity guarantees that this minimum is unique. Consequently, $a^*(s)$ is the unique global minimizer of h.

Because the minimization is over the whole space $\mathbb{R}^m$ (no constraints), the first-order necessary condition for optimality of a smooth, unconstrained objective function must hold:

$$\nabla h(a)\big|_{a=a^*(s)} = \nabla_a \rho(a)\big|_{a=a^*(s)} + g(s)^\top p = 0.$$

Strict convexity of ρ implies that the Hessian $\nabla^2 h(a) = \nabla^2 \rho(a)$ is positive definite everywhere; therefore the first-order condition is also sufficient for global optimality. This establishes (30) for almost every s. □

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Lemma 2. (Sufficient condition for optimality). Let Assumptions 1–2 hold. Let V be a locally Lipschitz function and suppose there exist measurable functions $a^* : \mathbb{R}^n \to \mathbb{A}$ and $p : \mathbb{R}^n \to \mathbb{R}^n$ with $p(s) \in \partial_C V(s)$ such that for almost every $s \in \mathbb{R}^n$ the following three conditions hold:

(i) Stationarity: $\nabla_a \rho(a)|_{a=a^*(s)} + g(s)^\top p(s) = 0$;

(ii) Optimality of the Hamiltonian:

$$\inf_{a\in\mathbb{A}} \big\{c(s,a) + p(s)\cdot(f(s)+g(s)a)\big\} = 0,$$

and $a^*(s)$ attains the infimum;

(iii) Viscosity supersolution: for every $\tilde{p} \in \partial_C V(s)$,

$$\inf_{a\in\mathbb{A}} \big\{c(s,a) + \tilde{p}\cdot(f(s)+g(s)a)\big\} \geq 0.$$

Then V is the optimal value function for the control problem with dynamics $\dot{s} = f(s) + g(s)a$ and running cost c, and a^* is an optimal feedback policy.

Proof. Conditions (ii) and (iii) together state that for almost every s:

- there exists $p(s) \in \partial_C V(s)$ for which

$$\inf_{a\in\mathbb{A}} \big\{c(s,a) + p(s)\cdot(f(s)+g(s)a)\big\} = 0,$$

and the infimum is realized by $a^*(s)$ (condition (ii));

- for every $\tilde{p} \in \partial_C V(s)$,

$$\inf_{a\in\mathbb{A}} \big\{c(s,a) + \tilde{p}\cdot(f(s)+g(s)a)\big\} \geq 0$$

(condition (iii)).

Thus V satisfies precisely the viscosity HJB condition of Definition 1: the subsolution property holds with equality for the particular generalized gradient $p(s)$, and the supersolution property is exactly condition (iii). Moreover, condition (ii) guarantees that a measurable optimal feedback a^* exists and that the closed-loop system admits solutions (the well-posedness of the feedback is ensured when a^* is discontinuous by interpreting solutions in the sense of Filippov; see Lemma 3).

Because V satisfies the viscosity HJB equation, it is a viscosity solution of the Hamilton–Jacobi–Bellman equation associated with the control problem. Under Assumptions 1–2, the classical uniqueness results for viscosity solutions [18, 25] imply that V is the unique viscosity solution. Since the true optimal value function V^* is also a viscosity solution of the same HJB equation (see, e.g., [18]), we must have $V = V^*$. Therefore V is the optimal value function.

Condition (i) (stationarity) is automatically fulfilled because $a^*(s)$ minimizes the Hamiltonian over $\mathbb{R}^m$ and the control cost ρ is smooth and strictly convex; it is stated separately for clarity and for use in constructive procedures. The policy a^* that attains the minimum is thus optimal. □

Remark. When V is continuously differentiable (C^1), we have $\partial_C V(s) = \{\nabla V(s)\}$, and the viscosity conditions reduce to the classical HJB equation. In this case, Lemma 1 recovers the standard first-order optimality condition, and Lemma 2 becomes the classical verification theorem. Thus, the non-smooth formulation strictly generalizes the smooth case.

Lemma 3. (Measurability of the optimal feedback.) Let Assumptions 1–2 hold and suppose a measurable selection $p(s) \in \partial_C V(s)$ is given.

(i) The set-valued map

$$\Phi(s) \coloneqq \arg\min_{a\in\mathbb{A}} \big\{c(s,a) + p(s)\cdot(f(s)+g(s)a)\big\}$$

is nonempty, compact, and measurable (its graph is Borel).

(ii) $\Phi(s)$ admits a measurable selection $a^* : \mathbb{R}^n \to \mathbb{A}$; i.e., there exists a Lebesgue-measurable function a^* such that $a^*(s) \in \Phi(s)$ for almost every s.

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(iii) The closed-loop vector field $\dot{s} = f(s) + g(s)a^*(s)$ is measurable and locally bounded, and therefore admits Filippov solutions [26] for any initial condition.

Proof. **(i) Non-emptiness, compactness, and measurability of $\Phi(s)$.** Fix $s \in \mathbb{R}^n$. The function

$$\psi_s(a) \coloneqq c(s,a) + p(s) \cdot \big(f(s) + g(s)a\big)$$

is continuous in a, because c is continuous (Assumption 1(c)), f and g are locally Lipschitz, and the inner product with $p(s)$ is linear. The control set $\mathbb{A}$ is compact and non-empty (Assumption 1(a)). By the Extreme Value Theorem, ψ_s attains its minimum on $\mathbb{A}$. Hence, $\Phi(s)$ is non-empty.

The set $\Phi(s) = \psi_s^{-1}(\{\min \psi_s\})$ is the inverse image of the closed singleton $\{\min \psi_s\}$ under a continuous map ψ_s; therefore it is closed. A closed subset of a compact is compact, so $\Phi(s)$ is compact.

We now show that Φ is measurable (i.e., its graph is a Borel subset of $\mathbb{R}^n \times \mathbb{A}$). For each fixed $a \in \mathbb{A}$, the map $s \mapsto \psi_s(a)$ is Lebesgue-measurable. Indeed, c is continuous, f and g are locally Lipschitz (hence measurable), and $p(s)$ is measurable by hypothesis; the sum of compositions of measurable functions is measurable. Thus $(s,a) \mapsto \psi_s(a)$ is a Carathéodory function (measurable in s for fixed a, and continuous in a for fixed s). For a Carathéodory objective and a constant compact constraint set, the argmin multifunction is measurable; its graph is a Borel subset of the product space $\mathbb{R}^n \times \mathbb{A}$ [27].

(ii) Existence of a measurable selection. From part (i), the correspondence $\Phi(s)$ has non-empty compact (hence closed) values and its graph $\mathrm{Gr}(\Phi)$ is a Borel subset of $\mathbb{R}^n \times \mathbb{A}$. The spaces $\mathbb{R}^n$ and $\mathbb{A} \subset \mathbb{R}^m$ are Polish (complete separable metric spaces). For a correspondence with closed values taking values in a Polish space, the property that its graph is a Borel set is equivalent to weak measurability [27]. Hence, $\Phi(s)$ is weakly measurable. The Kuratowski–Ryll-Nardzewski measurable selection theorem states that every weakly measurable set-valued map with non-empty closed values in a Polish space admits a Borel-measurable selection. Specifically, if $\mathrm{Gr}(\Phi)$ is measurable (which it is, being Borel-measurable) and $\Phi(s)$ is non-empty and closed for all s, then there exists a Borel-measurable function $a^* : \mathbb{R}^n \to \mathbb{A}$ such that $a^*(s) \in \Phi(s)$ for almost every s. Since every Borel-measurable function is Lebesgue-measurable, we obtain the required measurable selection, which we take as the optimal feedback.

(iii) Filippov solutions of the closed-loop system. We define the closed-loop vector field $F(s) := f(s) + g(s)a^*(s)$. The functions f and g are continuous (so Borel-measurable) and the function a^* is Lebesgue-measurable. The composition of continuous and measurable functions is measurable, so F is Lebesgue-measurable. The functions f, g are locally Lipschitz, so they are locally bounded. The action set $\mathbb{A}$ is compact by assumption, and therefore $a^*(s)$ stays within a bounded set. Hence for any bounded neighborhood of a state, the product $g(s)a^*(s)$ is bounded, and $F(s)$ is bounded. Thus F is locally bounded.

For a vector field that is measurable and locally bounded, Filippov's theorem [26] states that for every initial condition $s_0 \in \mathbb{R}^n$ there exists an absolutely continuous trajectory $s(t)$ defined on some interval $[0, T]$ such that

$$\dot{s}(t) \in K[F](s(t)) \quad \text{for almost all } t,$$

where $K[F]$ denotes the Filippov set. Thus the closed-loop system admits a solution in the sense of Filippov. If a^* is continuous, the Filippov set reduces to $\{F(s)\}$ and the trajectory is a classical Carathéodory solution. □

Theorem 1 (Non-smooth continuous-time deterministic converse optimality). Let Assumptions 1–2 hold, and let an input-coupling matrix $g : \mathbb{R}^n \to \mathbb{R}^{n\times m}$, a stage cost $c(s,a) = q(s) + \rho(a)$ with q continuous and $\rho \in C^2$ strictly convex, and a candidate locally Lipschitz value function $V : \mathbb{R}^n \to \mathbb{R}$ be given. Then the following statements are equivalent:

(a) There exists a drift $f : \mathbb{R}^n \to \mathbb{R}^n$ such that V is the optimal value function for the control-affine system $\dot{s} = f(s) + g(s)a$ with running cost c.

(b) There exist a drift $f : \mathbb{R}^n \to \mathbb{R}^n$, a measurable policy $a^* : \mathbb{R}^n \to \mathbb{A}$, and a measurable selection $p : \mathbb{R}^n \to \mathbb{R}^n$ with $p(s) \in \partial_C V(s)$ such that for every $s \in \mathbb{R}^n$ the following three conditions hold:

(i) $\nabla_a \rho(a)\big|_{a=a^*(s)} + g(s)^\top p(s) = 0$;

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(ii) $\inf_{a\in\mathbb{A}} \{c(s,a) + p(s)\cdot\big(f(s)+g(s)a\big)\} = 0$, and $a^*(s)$ attains the infimum;

(iii) for every $\tilde{p} \in \partial_C V(s)$, $\inf_{a\in\mathbb{A}} \{c(s,a) + \tilde{p}\cdot\big(f(s)+g(s)a\big)\} \geq 0$.

Whenever (b) holds, the policy a^* is optimal and V is the optimal value function.

Proof. (b) ⇒ (a) – Sufficiency. If statement (b) holds, we are given a drift f, a measurable policy a^*, and a measurable selection $p(s) \in \partial_C V(s)$ satisfying conditions (i)–(iii). These are exactly the hypotheses of Lemma 2 (sufficient condition for optimality). Lemma 2 therefore applies and guarantees that, for the control-affine system $\dot{s} = f(s) + g(s)a$ with running cost c, the function V is the optimal value function and a^* is an optimal feedback policy. Hence statement (a) is true.

(a) ⇒ (b) – Necessity. Assume that (a) holds, i.e., there exists a drift f_0 such that V is the optimal value function for the system $\dot{s} = f_0(s) + g(s)a$ with cost c. Because the control problem satisfies Assumptions 1–2, the optimal value function is a viscosity solution of the associated Hamilton–Jacobi–Bellman equation [18, 25]. Hence the triple (f_0, g, c) satisfies the viscosity HJB condition of Definition 1: there exist a measurable policy $a^* : \mathbb{R}^n \to \mathbb{A}$ and a measurable selection $p(s) \in \partial_C V(s)$ such that for almost every s

$$\inf_{a\in\mathbb{A}} \{c(s,a) + p(s)\cdot(f_0(s)+g(s)a)\} = 0, \qquad a^*(s) \text{ attains the infimum,}$$

and for every $\tilde{p} \in \partial_C V(s)$,

$$\inf_{a\in\mathbb{A}} \{c(s,a) + \tilde{p}\cdot(f_0(s)+g(s)a)\} \geq 0.$$

Thus conditions (ii) and (iii) of statement (b) hold with the drift $f = f_0$.

It remains to verify the stationarity condition (i). The hypotheses of Lemma 1 (first-order optimality) are fulfilled: V is a viscosity solution, the control cost ρ is strictly convex and C^2, and the selections a^*, p achieve the zero infimum. Lemma 1 therefore yields

$$\nabla_a \rho(a)\big|_{a=a^*(s)} + g(s)^\top p(s) = 0,$$

which is precisely condition (i) of statement (b). Consequently, the same drift f_0 together with the selections a^* and p satisfies all three requirements of (b). This completes the proof of the equivalence. □

Remark on the role of Lemma 3. Although Lemma 3 is not explicitly invoked in the proof of Theorem 1, it provides the essential technical foundation that makes the entire converse optimality framework well-posed. The viscosity HJB condition (Definition 1) and the sufficient conditions in Lemma 2 both require the existence of a measurable optimal feedback a^*. Lemma 3 demonstrates that, under Assumptions 1–2, the pointwise minimisation of the Hamiltonian yields a set-valued map that is non-empty, compact, and weakly measurable, and therefore admits a measurable selector. It further guarantees that the resulting closed-loop vector field $f(s) + g(s)a^*(s)$ is measurable and locally bounded, so that solutions exist in the Filippov sense even when the optimal policy is discontinuous. Consequently, Lemma 3 justifies the use of measurable feedback policies throughout the paper and ensures that the benchmark systems generated via converse optimality are mathematically sound and can be simulated in practice.

G.4 General Constructive Procedure

We now formulate a systematic procedure that, given a candidate locally Lipschitz value function V, produces a control-affine system and a running cost for which V is optimal in the viscosity sense. The procedure does not assume any specific form for the control cost ρ beyond strict convexity and C^2 regularity. Quadratic costs are treated in the next subsection.

We work under Assumptions 1–2 of Section 3. The input-coupling matrix $g : \mathbb{R}^n \to \mathbb{R}^{n\times m}$ is prescribed and satisfies Assumption 1(b). The control set $\mathbb{A}$ is compact and non-empty (Assumption 1(a)); in the unconstrained case we take $\mathbb{A} = \mathbb{R}^m$. The candidate value function $V : \mathbb{R}^n \to \mathbb{R}$ is locally Lipschitz (Assumption 2) and its Clarke subdifferential $\partial_C V(s)$ is assumed to be explicitly characterized.

The procedure consists of six steps.

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Step 1. **Subdifferential analysis.** Compute the Clarke subdifferential $\partial_C V(s)$ at every state $s \in \mathbb{R}^n$. Identify:

- smooth regions – open sets where $\partial_C V(s) = \{\nabla V(s)\}$ is a singleton;
- switching surfaces – lower-dimensional manifolds where V is non-differentiable and $\partial_C V(s)$ contains more than one element.

On each switching surface, express $\partial_C V(s)$ as the convex hull of a finite collection of limiting gradients from adjacent smooth regions. Introduce local parameters $\alpha \in [0,1]^k$ (a single scalar α if the surface is of codimension one) that parametrize this convex hull.

Step 2. **Selection of a generalized gradient.** Choose a measurable selection $p : \mathbb{R}^n \to \mathbb{R}^n$ such that $p(s) \in \partial_C V(s)$ for every s. In smooth regions the selection is unique. On switching surfaces the selection may depend on the parameters α introduced in Step 1; different choices of α yield different admissible systems that all share the same value function V.

Step 3. **Construction of the drift.** Choose a (possibly state-dependent) positive definite gain matrix $K(s) \succ 0$ that is locally Lipschitz and define the drift

$$f(s) \coloneqq -K(s)\, p(s). \tag{31}$$

The negative sign ensures that the uncontrolled dynamics $\dot{s} = f(s)$ drives the state in the direction of steepest descent of V (generalized gradient descent). This choice guarantees

$$p(s)\cdot f(s) = -\,p(s)^\top K(s) p(s) \le 0,$$

which will be used to enforce non-negativity of the state cost.

Step 4. **Determination of the state cost from the Hamiltonian condition.** With the drift fixed by (31), the optimal feedback a^* is determined by the stationarity condition of Theorem 1 (b). For a general strictly convex $\rho \in C^2$, the equation

$$\nabla_a \rho(a)\big|_{a=a^*(s)} + g(s)^\top p(s) = 0 \tag{32}$$

implicitly defines $a^*(s)$. Because ρ is strictly convex and coercive, the solution $a^*(s)$ exists, is unique, and is a measurable function of s (by the measurable selection theorem, Lemma 3).

The state cost $q(s)$ is then obtained from the Hamiltonian optimality condition (condition (ii) of Theorem 1), which for the chosen f and a^* reads

$$0 = q(s) + \rho(a^*(s)) + p(s)\cdot\big(f(s) + g(s)a^*(s)\big). \tag{33}$$

Solving (33) for $q(s)$ gives

$$q(s) = -\rho(a^*(s)) - p(s)\cdot\big(f(s) + g(s)a^*(s)\big). \tag{34}$$

With $f(s) = -K(s)p(s)$ the term $p(s)\cdot f(s) = -p(s)^\top K(s)p(s) \le 0$. The sign of the remaining term depends on the specific control cost. In practice, $q(s)$ is computed directly from (34) and then checked for non-negativity.

Step 5. **Verification of the viscosity conditions.** With f, a^* and q constructed as above, verify the viscosity HJB condition of Definition 1:

- *Subsolution property.* By construction, for the selected $p(s) \in \partial_C V(s)$,

$$\inf_{a\in\mathbb{A}} \big\{c(s,a) + p(s)\cdot(f(s) + g(s)a)\big\} = 0,$$

and $a^*(s)$ attains the infimum. This follows directly from (32) and the definition of $q(s)$.

- *Supersolution property.* For every $\tilde{p} \in \partial_C V(s)$ we must have

$$\inf_{a\in\mathbb{A}} \big\{c(s,a) + \tilde{p}\cdot(f(s) + g(s)a)\big\} \ge 0.$$

This inequality is not automatically satisfied; it imposes a constraint on the gain matrix $K(s)$ and on the geometry of the input-coupling $g(s)$. It must be checked for the specific V and the chosen $K(s)$. Sufficient conditions are discussed in the quadratic case in Subsection 5.2 and can be adapted to general costs.

If the supersolution condition fails for the chosen $K(s)$, one may increase $K(s)$ and recompute $q(s)$ until the inequality holds. Theorem 1 guarantees that if such a $K(s)$ exists, the constructed system is optimal.

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Step 6. **Well-posedness of the closed-loop system.** The optimal feedback $a^*(s)$ obtained from (32) may be discontinuous on switching surfaces. The closed-loop dynamics

$$\dot{s} = f(s) + g(s)a^*(s)$$

must therefore be interpreted as a differential inclusion in the sense of Filippov [26]. Lemma 3 guarantees that the right-hand side is measurable and locally bounded, so that absolutely continuous Filippov solutions exist for any initial condition. This ensures the well-posedness of the constructed control problem.

The procedure yields a triple (f, g, c) that satisfies all hypotheses of Theorem 1, thereby guaranteeing that V is the optimal value function and a^* is an optimal feedback.

G.5 Construction with a Quadratic Control Cost

We now specialize the general procedure of Subsection G.4 to the practically important case of a quadratic control cost. This is the setting that will be used for the benchmark generation examples described later.

Let the control cost be

$$\rho(a) = \tfrac{1}{2}\, a^\top R\, a, \qquad R \succ 0,$$

and assume that the control space is unconstrained, i.e. $\mathbb{A} = \mathbb{R}^m$. (The treatment of compact control sets can be handled by projection of the optimal action; see the remark at the end of this subsection.) The drift is chosen exactly as in the general case,

$$f(s) = -K(s)\, p(s), \qquad K(s) \succ 0, \tag{35}$$

where $p(s)$ is the measurable selection from the Clarke subdifferential $\partial_C V(s)$ produced in Step 2. Because ρ is quadratic, the stationarity condition

$$\nabla\rho(a) + g^\top p = Ra + g^\top p = 0$$

can be solved explicitly for the optimal action:

$$a^*(s) = -R^{-1}\, g(s)^\top\, p(s). \tag{36}$$

From the Hamiltonian optimality condition (33) we have

$$0 = q(s) + \tfrac{1}{2}\, a^*(s)^\top R\, a^*(s) + p(s)\cdot\big(f(s) + g(s)a^*(s)\big). \tag{37}$$

Substituting (35) and (36) and solving for $q(s)$ gives a closed-form expression for the state cost:

$$\begin{aligned} q(s) &= -\tfrac{1}{2}\, a^*(s)^\top R\, a^*(s) - p(s)\cdot\big(f(s) + g(s)a^*(s)\big) \\ &= \tfrac{1}{2}\, p(s)^\top g(s) R^{-1} g(s)^\top p(s) + p(s)^\top K(s)\, p(s). \end{aligned} \tag{38}$$

The right-hand side is manifestly non-negative because $R^{-1} \succ 0$ and $K(s) \succ 0$; thus the state cost automatically satisfies the positivity requirement.

Automatic satisfaction of conditions (i) and (ii). With the choices (35), (36) and (38), condition (i) (stationarity) holds by construction. Condition (ii) (optimality of the Hamiltonian) is also satisfied, because for the selected generalized gradient $p(s)$,

$$\inf_{a\in\mathbb{R}^m} \big\{c(s,a) + p(s)\cdot(f(s) + g(s)a)\big\} = c(s, a^*(s)) + p(s)\cdot(f(s) + g(s)a^*(s)) = 0,$$

as a direct algebraic verification confirms.

Supersolution condition. For fixed s and $\tilde{p}$, let

$$\kappa(a) = q(s) + \frac{1}{2} a^\top R a + \tilde{p}(s)\cdot(f(s) + g(s)a).$$

This expression is quadratic in a. The only remaining requirement is the supersolution inequality (condition (iii) of Lemma 2), which for the quadratic cost reads:

$$\forall \tilde{p} \in \partial_C V(s) : \inf_{a\in\mathbb{R}^m} \kappa(a) = \inf_{a\in\mathbb{R}^m} \big\{\tfrac{1}{2} a^\top R a + q(s) + \tilde{p}\cdot\big(f(s) + g(s)a\big)\big\} \geq 0. \tag{39}$$

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We now compute its infimum over a. For a fixed $\tilde{p}$, the gradient with respect to a is $Ra + g(s)^\top \tilde{p}(s)$. Setting it to zero yields the unique minimizer:

$$a^*_{\tilde{p}} = -R^{-1} g(s)^\top \tilde{p}(s).$$

The infimum is obtained by substituting $a^*_{\tilde{p}}$ into κ:

$$\inf_a \kappa(a) = \kappa(a^*_{\tilde{p}}) = q(s) + \frac{1}{2}(a^*_{\tilde{p}})^\top R a^*_{\tilde{p}} + \tilde{p}(s) \cdot (f(s) + g(s) a^*_{\tilde{p}}).$$

We now simplify the terms involving $a^*_{\tilde{p}}$:

$$\frac{1}{2}(a^*_{\tilde{p}})^\top R a^*_{\tilde{p}} = \frac{1}{2}(-R^{-1} g(s)^\top \tilde{p}(s))^\top R (-R^{-1} g(s)^\top \tilde{p}(s)) = \frac{1}{2} \tilde{p}(s)^\top g(s) R^{-1} g(s)^\top \tilde{p}(s),$$

$$\tilde{p}(s) \cdot g(s) a^*_{\tilde{p}} = \tilde{p}(s) \cdot (-g(s) R^{-1} g(s)^\top \tilde{p}(s)).$$

Summing these two yields:

$$\frac{1}{2}(a^*_{\tilde{p}})^\top R a^*_{\tilde{p}} + \tilde{p} \cdot g(s) a^*_{\tilde{p}} = \frac{1}{2} \tilde{p}^\top g(s) R^{-1} g(s)^\top \tilde{p} - \tilde{p}^\top g(s) R^{-1} g(s)^\top \tilde{p} = -\frac{1}{2} \tilde{p}^\top g(s) R^{-1} g(s)^\top \tilde{p}.$$

We now substitute $f(s) = -K(s)p(s)$ and $q(s)$ from (38), which gives:

$$\begin{aligned} \inf_a \kappa(a) = \Big(\tfrac{1}{2}\, p(s)^\top g(s) R^{-1} g(s)^\top p(s) + p(s)^\top K(s)\, p(s) \Big) \\ - \tfrac{1}{2}\, \tilde{p}(s)^\top g(s) R^{-1} g(s)^\top \tilde{p}(s) - \tilde{p}(s)^\top K(s)\, p(s) \geq 0. \end{aligned}$$

This inequality must be checked for the chosen gain matrix $K(s)$. If $\partial_C V(s)$ is a singleton (smooth region), then $\tilde{p} = p(s)$ and the inequality holds with equality for any $K(s)$. However, on switching surfaces, where $\partial_C V(s)$ contains multiple generalized gradients, condition (39) imposes an additional constraint. Indeed, on a switching surface, $\partial_C V(s)$ contains other generalized gradients. For any $\tilde{p} \neq p$ the inequality is not automatically satisfied. Consider

$$p^\top K p - \tilde{p}^\top K p.$$

This can be negative, for example, if $\tilde{p}$ is larger than p in some direction. The remaining $p^\top K p - \tilde{p}^\top K p = k\|p\|^2 - k\tilde{p} \cdot p$ terms of the inequality can be either positive or negative. The inequality therefore forbids certain combinations of $\tilde{p}$ and p unless K is chosen appropriately. In other words, it is a condition that couples the choice of the gain matrix K to the geometry of the Clarke subdifferential. Without checking it, we cannot guarantee optimality.

Now, suppose $K = kI$, with a scalar $k > 0$. Then (39) becomes

$$k\big(\|p\|^2 - \tilde{p}^\top \cdot p\big) \geq \tfrac{1}{2}\, \tilde{p}^\top g R^{-1} g^\top \tilde{p} - \tfrac{1}{2}\, p^\top g R^{-1} g^\top p.$$

The right-hand side of this inequality does not depend on k. If $\|p\|^2 > \tilde{p}^\top \cdot p$ for all $\tilde{p} \in \partial_C V(s)$, then the left-hand side is linear in k with a positive coefficient, and choosing k sufficiently large guarantees the inequality. If $\|p\|^2 \leq \tilde{p}^\top \cdot p$ for some $\tilde{p}$, the coefficient of k becomes non-positive, and increasing k cannot help; one must then either select a different generalized gradient $p(s) \in \partial_C V(s)$ for which the condition holds, or use a state-dependent gain matrix $K(s)$ to satisfy the inequality.

When $\|p\|^2 > \tilde{p}^\top \cdot p$ for all $\tilde{p} \in \partial_C V(s)$, the worst-case analysis finds the smallest possible k (or more generally the smallest K in the positive-definite sense) that makes (39) hold for all $\tilde{p} \in \partial_C V(s)$. We require

$$\min_{\tilde{p} \in \partial_C V(s)} \big[p^\top K p - \tilde{p}^\top K p + \tfrac{1}{2}\, \tilde{p}^\top g R^{-1} g^\top \tilde{p} - \tfrac{1}{2}\, p^\top g R^{-1} g^\top p \big] \geq 0.$$

The minimum is attained because $\partial_C V(s)$ is compact. Setting $K = kI$ and rearranging, this becomes

$$k\Big(\min_{\tilde{p} \in \partial_C V(s)} \big\{ \|p\|^2 - \tilde{p} \cdot p \big\} \Big) \;\geq\; \max_{\tilde{p} \in \partial_C V(s)} \big\{ \tfrac{1}{2}\, \tilde{p}^\top g R^{-1} g^\top \tilde{p} \big\} - \tfrac{1}{2}\, p^\top g R^{-1} g^\top p.$$

If the coefficient of k is positive, we can divide to obtain an explicit lower bound

$$k \;\geq\; \frac{\max_{\tilde{p} \in \partial_C V(s)} \big\{ \tfrac{1}{2}\, \tilde{p}^\top g R^{-1} g^\top \tilde{p} \big\} - \tfrac{1}{2}\, p^\top g R^{-1} g^\top p}{\min_{\tilde{p} \in \partial_C V(s)} \big\{ \|p\|^2 - \tilde{p} \cdot p \big\}},$$

assuming the denominator is positive.

This is a stability result. If the chosen gain matrix is too small, the supersolution might fail; but when $\|p\|^2 > \tilde{p}^\top \cdot p$ for all $\tilde{p} \in \partial_C V(s)$, increasing k always eventually works (for compact subdifferentials). The worst-case analysis tells how large it must be.

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G.6 Example: Piecewise-Quadratic Value Function

We apply the construction procedure to a two-dimensional piecewise-quadratic value function. This example shows how the non-smooth converse optimality framework works. We consider a control-affine system with dynamics

$$\dot{s} = f(s) + a, \quad a \in \mathbb{R}^2.$$

Here, we set $g(s) \equiv I$. This choice is made to focus on the handling of non-smooth value functions and the construction of converse optimality in the viscosity sense. We avoid the algebraic complexity introduced by a state-dependent input matrix. The extension to general $g(s)$ is discussed afterwards.

Let the state be $s = (x, y)^\top \in \mathbb{R}^2$ and let the candidate value function be

$$V(s) = \begin{cases} x^2 + 2xy + 2y^2, & x \geq 0, \\ 2x^2 - 2xy + 2y^2, & x < 0, \end{cases}$$

This function is continuous, locally Lipschitz, convex, and differentiable everywhere except on the line $x = 0$. The cost is taken quadratic in control:

$$c(s, a) = q(s) + \frac{1}{2}\|a\|^2.$$

Step 1. We compute the Clarke subdifferential of V:

- For $x > 0$: $\partial_C V(s) = \{(2x + 2y,\ 2x + 4y)^\top\}$
- For $x < 0$: $\partial_C V(s) = \{(4x - 2y,\ -2x + 4y)^\top\}$
- For $x = 0$: $\partial_C V(0, y) = \mathrm{co}\{(2y, 4y)^\top, (-2y, 4y)^\top\} = \{(2(2\alpha - 1)y,\ 4y)^\top \mid \alpha \in [0, 1]\}$ (convex hull)

The two extreme cases are $\alpha = 1$ (the right-hand gradient $(2y, 4y)^\top$) and $\alpha = 0$ (the left-hand gradient $(-2y, 4y)^\top$).

Step 2. We select a generalized gradient. We choose the right-hand limiting gradient on the switching line,

$$p(s) = \begin{cases} (2x + 2y, 2x + 4y)^\top, & x > 0, \\ (4x - 2y, -2x + 4y)^\top, & x < 0, \\ (2y, 4y)^\top, & x = 0. \end{cases}$$

This corresponds to $\alpha = 1$ in the convex-hull parametrization on $x = 0$; other choices of α will be explored later.

Step 3. We choose a constant gain matrix $K(s) = kI$ with $k > 0$. By the general formula (35) the drift is

$$f(s) = -k\, p(s).$$

For simplicity we take $k = 1$ in all numerical illustrations, but the physical meaning of k is interesting. The function V is a Lyapunov function for the uncontrolled dynamics $\dot{s} = f(s)$: indeed $\dot{V}(s) = p \cdot f = -k\|p\|^2 \leq 0$ for any $p \in \partial_C V(s)$, so V decreases along trajectories. The choice $f = -kp$ corresponds to a generalized gradient descent; the factor k quantifies how much of the stabilization is performed by the natural dynamics rather than by the control. When $k \to 0$, all the stabilization task is put on the control a, which makes the task more difficult; when $k \gg 1$ the open-loop system already drives the state rapidly toward the origin. The system itself tends to equilibrium, without needing much help from the control. Hence k is an additional degree of freedom that can be used to create benchmark families with different levels of difficulty. With $k = 1$ the drift becomes

$$f(s) = \begin{cases} -(2x + 2y,\ 2x + 4y)^\top, & x > 0, \\ -(4x - 2y,\ -2x + 4y)^\top, & x < 0. \end{cases}$$

On the switching line we must decide how to define f. On the line $x = 0$, the subdifferential is a segment. Continuity of f across $x = 0$ cannot be achieved unless $y = 0$, because the limits from the right and left differ. From the right, ($x > 0$), we have

$$f_{\text{right}}(x, y) = -(2x + 2y,\ 2x + 4y)^\top \quad \text{and} \quad \lim_{x \to 0^+} f_{\text{right}}(x, y) = -(2y, 4y)^\top.$$

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From the left ($x < 0$) we have

$$f_{\text{left}}(x, y) = -(4x - 2y,\ -2x + 4y)^\top \quad \text{and} \quad \lim_{x \to 0^-} f_{\text{left}}(x, y) = (2y, -4y)^\top.$$

We therefore define f on $x = 0$ by the right-hand limit

$$f(0, y) = \lim_{x \to 0^+} f(s) = -(2y, 4y)^\top.$$

This choice makes f right-continuous and yields a drift that is consistent with the selected generalized gradient on the switching surface. This is useful because we will see that the optimal control will have a similar jump on the line $x = 0$.

Step 4. We now compute the optimal feedback and the state cost. For any $p \in \partial_C V(s)$, the viscosity HJB condition requires

$$\inf_{a \in \mathbb{A}} \{c(s, a) + p \cdot (f(s) + a)\} = 0. \tag{40}$$

The minimizer for the quadratic cost is $a^*(s, p) = -p$ (assuming p lies in $\mathbb{A}$). Indeed, we begin with the Hamiltonian:

$$H(s, a, p) = c(s, a) + p \cdot (f(s) + a).$$

Substituting the quadratic control cost $c(s, a) = q(s) + \frac{1}{2}\|a\|^2$ yields:

$$H(s, a, p) = q(s) + \frac{1}{2}\|a\|^2 + p \cdot f(s) + p \cdot a.$$

To satisfy the HJB condition, we minimize H with respect to a:

$$a^*(s, p) = \arg\min_{a \in \mathbb{A}} \left\{ \frac{1}{2}\|a\|^2 + p \cdot a \right\},$$

since the terms $q(s)$ and $p \cdot f(s)$ are independent of a. When there are no control constraints, the quadratic form can be completed:

$$\frac{1}{2}\|a\|^2 + p \cdot a = \frac{1}{2}\|a + p\|^2 - \frac{1}{2}\|p\|^2.$$

Setting the gradient with respect to a to zero:

$$\nabla_a \left(\frac{1}{2}\|a\|^2 + p \cdot a \right) = a + p = 0,$$

we find that the minimum is attained at

$$a^*(s, p) = -p.$$

Substituting into (40) gives the state cost

$$q(s) + \frac{1}{2}\|p\|^2 + p \cdot f(s) - \|p\|^2 = 0 \Longrightarrow q(s) = \frac{1}{2}\|p\|^2 - p \cdot f(s) = \left(\frac{1}{2} + k\right)\|p\|^2 \geq 0. \tag{41}$$

For $k = 1$, this gives $q(s) = \frac{3}{2}\|p(s)\|^2 \geq 0$ without additional constraints. All of this yields a stabilizing dynamics. Indeed, we have:

- $f(x, y) = -(2x + 2y,\ 2x + 4y)^\top = -P_1(x, y)^\top, \quad x > 0,$
- $f(x, y) = -(4x - 2y,\ -2x + 4y)^\top = -P_2(x, y)^\top, \quad x < 0,$

where

$$P_1 = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}, \qquad P_2 = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$$

This system is thus asymptotically stable, because $-P_i$ are Hurwitz matrices. Also, conditions (i) and (ii) of Lemma 2 are satisfied.

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Step 5. We now verify the viscosity conditions. In the smooth regions $x \neq 0$ the Clarke subdifferential is a singleton and the classical HJB equation holds by construction. On the switching line $x = 0$ we must verify the full viscosity HJB condition of Definition 1 for the constructed system.

Subsolution. For the selected generalized gradient $p(0, y) = (2y, 4y)^\top$ the Hamiltonian evaluates to zero:

$$\begin{aligned} c((0,y),a^*(0,y)) + p(0,y)\cdot(f(0,y)+a^*(0,y)) \\ &= q(0,y) + \frac{1}{2}\|a^*\|^2 + (2y,4y)\cdot\big(-(2y,4y)^\top - (2y,4y)^\top\big) \\ &= \frac{3}{2}(4y^2+16y^2) + \frac{1}{2}(4y^2+16y^2) + (2y,4y)\cdot(-4y,-8y)^\top \\ &= \frac{3}{2}\cdot 20y^2 + \frac{1}{2}\cdot 20y^2 - (8y^2+32y^2) \\ &= 30y^2 + 10y^2 - 40y^2 = 0. \end{aligned}$$

Thus the subsolution property holds with equality.

Supersolution. We must show that for every $\tilde{p} \in \partial_C V(0, y)$,

$$\inf_{a\in\mathbb{R}^2}\big\{c((0,y),a) + \tilde{p}\cdot(f(0,y)+ga)\big\} \geq 0.$$

Substituting the expressions for f, q and $g = I$, $R = I$ and using the general inequality from Subsection G.5 (equation (39)) reduces the condition to

$$-\frac{1}{2}\|\tilde{p}\|^2 + \frac{1}{2}\|p\|^2 + \|p\|^2 - \tilde{p}\cdot p \geq 0,$$

where $p = (2y, 4y)^\top$ and $\tilde{p} = (2(2\alpha-1)y,\ 4y)^\top$ with $\alpha \in [0, 1]$. After simplification this becomes

$$8(1-\alpha^2)y^2 \ \geq \ 0.$$

The quadratic $8(1 - \alpha^2)y^2$ is non-negative on $[0, 1]$, attaining its minimum 0 at $\alpha = 1$. Hence the supersolution inequality holds for all $\alpha \in [0, 1]$ and all y.

Therefore the viscosity HJB condition is satisfied, and Theorem 1 guarantees that V is the optimal value function for the drift f and a^* is an optimal feedback policy.

Homotopy parameterization. The parameter $\alpha \in [0, 1]$ that parametrizes the Clarke subdifferential on the switching line can be used to generate a family of admissible systems. For any fixed $\alpha \in [0, 1]$, choose on $x = 0$

$$p_\alpha(0,y) = \big(2(2\alpha-1)y,\ 4y\big)^\top,$$

and modify the drift and state cost on the switching line accordingly:

$$f_\alpha(0,y) = -k\,p_\alpha(0,y), \qquad q_\alpha(0,y) = \Big(\frac{1}{2}+k\Big)\big(4(2\alpha-1)^2+16\big)y^2.$$

The drift and cost in the smooth regions remain unchanged. Varying α yields a parameterized family $\{(f_\alpha, q_\alpha, a^*_\alpha)\}_{\alpha\in[0,1]}$ of control problems that all share the same optimal value function V. The limiting cases $\alpha = 1$ and $\alpha = 0$ correspond, respectively, to the right-continuous and left-continuous drifts; intermediate values interpolate between the two.

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Explicit piecewise formulas (for $k = 1$, $\alpha = 1$). Collecting the expressions obtained above, the benchmark system is given by

$$f(s) = \begin{cases} -\begin{pmatrix} 2x + 2y \\ 2x + 4y \end{pmatrix}, & x \geq 0, \\ -\begin{pmatrix} 4x - 2y \\ -2x + 4y \end{pmatrix}, & x < 0, \end{cases} \qquad f(0, y) = -\begin{pmatrix} 2y \\ 4y \end{pmatrix},$$

$$a^*(s) = \begin{cases} -\begin{pmatrix} 2x + 2y \\ 2x + 4y \end{pmatrix}, & x > 0, \\ -\begin{pmatrix} 4x - 2y \\ -2x + 4y \end{pmatrix}, & x < 0, \\ -\begin{pmatrix} 2y \\ 4y \end{pmatrix}, & x = 0, \end{cases}$$

$$q(s) = \begin{cases} \frac{3}{2}\big((2x + 2y)^2 + (2x + 4y)^2\big), & x \geq 0, \\ \frac{3}{2}\big((4x - 2y)^2 + (-2x + 4y)^2\big), & x < 0, \end{cases} \qquad c(s, a) = q(s) + \frac{1}{2}\|a\|^2.$$

G.7 Remarks on the Problem Geometry and Multiple Switching Surfaces

Geometric interpretation via Lie derivatives. In continuous-time control-affine systems, the quantity $g(s)^\top p(s)$ appearing in the optimal feedback (36) admits a natural geometric interpretation: it is a vector of Lie derivatives of V along the columns of the input-coupling matrix. Let the control vector fields $g_1(s), \ldots, g_m(s)$ be the columns of $g(s)$. Each g_i describes the instantaneous direction in which the state moves when the i-th control input is applied. For a smooth function V, the Lie derivative of V along the control vector field g_i is the directional derivative

$$\mathcal{L}_{g_i} V(s) = \nabla V(s) \cdot g_i(s).$$

It tells us how fast V changes if the state moves in the direction given by g_i. Hence, the vector $\mathcal{L}_g V(s) = (\mathcal{L}_{g_1} V(s), \ldots, \mathcal{L}_{g_m} V(s))^\top$ contains the rates of change of V in all actuated directions. In the non-smooth setting studied here, ∇V is replaced by an element of the Clarke subdifferential. In smooth continuous-time converse optimality, the optimal feedback is given in [14]:

$$a^*(s) = -R^{-1} \mathcal{L}_g V(s) = -R^{-1} g(s)^\top \nabla V(s).$$

This gradient is the steepest descent of V projected onto the subspace spanned by the columns of $g(s)$, and weighted by the inverse of the control-cost matrix R^{-1}.

With a measurable selection $p(s) \in \partial_C V(s)$, the scalar $p(s) \cdot g_i(s)$ is a selected Lie derivative from the set

$$\mathcal{L}_{g_i}^{\mathrm{Clarke}} V(s) \coloneqq \{\, \zeta \cdot g_i(s) : \zeta \in \partial_C V(s) \,\}.$$

This set is a closed interval in $\mathbb{R}$. Each coordinate of the control action vector is an element selected from each corresponding interval. The vector $g(s)^\top p(s) = (\mathcal{L}_{g_1}^{\mathrm{Clarke}} V(s), \ldots, \mathcal{L}_{g_m}^{\mathrm{Clarke}} V(s))^\top$ thus represents the chosen generalized rate of change of V in each actuated direction. The optimal feedback

$$a^*(s) = -R^{-1} g(s)^\top p(s) = -R^{-1} \mathcal{L}_g^{\mathrm{Clarke}} V(s)$$

steers the system opposite to the projection of the generalized gradient onto the control subspace. On smooth regions, the selection of the components of $g(s)^\top p(s)$ is unique and coincides with the classical gradient. On a switching surface, $\partial_C V(s)$ is the convex hull of limiting gradients from adjacent smooth regions. Hence, the set $\mathcal{L}_{g_i}^{\mathrm{Clarke}} V(s)$ is the convex hull of the classical Lie derivatives from the adjacent smooth regions. Its width quantifies the severity of the non-smoothness: in the piecewise-quadratic example, with $g = I$ we obtained $\mathcal{L}_{g_2}^{\mathrm{Clarke}} V(0, y) = [2y,\, 4y]$, an interval of length $2|y|$. The optimal policy selects a specific point from this interval, depending on the homotopy

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parameter α. Varying α continuously varies the strength of the optimal feedback on the switching line while preserving global optimality.

The supersolution condition of Lemma 2

$$\inf_{a\in\mathbb{A}} \left\{c(s,a) + \tilde{p}\cdot\left(f(s)+g(s)a\right)\right\} \geq 0 \quad \forall \tilde{p} \in \partial_C V(s)$$

can also be interpreted using Lie derivatives. For example, in the quadratic case, we obtain in (39)

$$\inf_{a\in\mathbb{R}^m} \left\{\tfrac{1}{2}a^\top Ra + q(s) + \tilde{p}\cdot\big(f(s)+g(s)a\big)\right\} \geq 0 \quad \forall \tilde{p} \in \partial_C V(s).$$

With Lie derivatives, we obtain

$$-\frac{1}{2}\|\mathcal{L}_g^{\tilde{p}}V\|^2_{R^{-1}} + \frac{1}{2}\|\mathcal{L}_g^{p}V\|^2_{R^{-1}} + \|p\|^2_K - \langle \tilde{p}, p\rangle_K \geq 0,$$

where $\mathcal{L}_g^{\tilde{p}}V = g^\top\tilde{p}$ and $\mathcal{L}_g^{p}V = g^\top p$ are respectively the tested and the selected Lie-derivative vectors, and $\|\cdot\|_{R^{-1}}$ and $\|\cdot\|_K$ are weighted norms. If we want the inequality to hold for all $\tilde{p}$, the selected Lie-derivative vector must not be too small in comparison to the other possible Lie derivatives of the Clarke subdifferential. It prevents the controller from being too soft on the switching surface.

Multiple switching surfaces and parameterization. When the candidate value function V has several switching surfaces (that can also intersect), the Clarke subdifferential at a point is the convex hull of limiting gradients from all adjacent smooth regions. The corresponding Lie-derivative set becomes a polytope, parametrized by a vector $\alpha \in [0,1]^k$ with k independent barycentric coordinates. The construction of Section G.5 then yields a k-parameter family of admissible systems $\{(f_\alpha, q_\alpha, a^*_\alpha)\}_{\alpha\in[0,1]^k}$, all sharing the same optimal value function V. Each component of α independently controls the sharpness of the non-smooth behavior along a particular switching surface, providing a multi-dimensional difficulty knob for benchmark generation.

G.8 Problem Definition and Generator Pipeline

Each problem instance is a non-smooth, continuous-time, deterministic, control-affine problem of the form:

$$\dot{s} = f(s) + g(s)a, \qquad s\in\mathbb{R}^n,\ a\in\mathbb{R}^m,$$

with running cost $c(s,a) = q(s) + \rho(a)$ where $\rho(a) = \frac{1}{2}a^\top Ra$ ($R \succ 0$) and $q(s)$ is given by (38). The drift f is constructed as $f(s) = -K(s)p(s)$ with a chosen gain matrix $K(s) \succ 0$ and a measurable selection $p(s) \in \partial_C V(s)$. The optimal feedback is $a^*(s) = -R^{-1}g(s)^\top p(s)$, and the triple (f, g, c) satisfies the viscosity HJB condition of Definition 1 by Theorem 1. The certified optimal value function is the prescribed V, and the optimal policy is $\pi^*(s) = a^*(s)$.

A benchmark family is a parametric collection of such control problems:

$$\mathcal{F}_\psi = \{M_\psi : \psi \in \Psi\},$$

where each M_ψ is defined by a tuple $(f_\psi, g_\psi, q_\psi, \rho, R)$. The parameter space Ψ encodes tunable difficulty knobs (dimension, control authority, nonlinearity strength, non-smoothness severity) while preserving the analytical optimal solution. The certified ground truth (π^*_ψ, V^*_ψ) follows directly from the construction.

The generator pipeline is summarized in Algorithm 2. It implements the six-step procedure of Section G.4 for the quadratic case (Subsection G.5) and exports validated YAML fixtures.

G.9 An Example: the Piecewise-Quadratic Family

As an example, among others, of a benchmark family that can be generated, consider the two-dimensional state $s = (x, y)^\top$ and a candidate value function

$$V(s) = \begin{cases} x^2 + 2\beta xy + 2y^2, & x \geq 0, \\ 2x^2 + 2\gamma xy + 2y^2, & x < 0, \end{cases} \qquad \beta,\gamma\in\mathbb{R},\ \beta\neq\gamma.$$

The parameters β and γ are the off-diagonal coefficients of the two quadratic forms. At a point on the switching line $(0, y)$, the gradient from the right is $(2\beta y, 4y)^\top$ and from the left is $(2\gamma y, 4y)^\top$.

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Algorithm 2 Converse-Optimal Benchmark Generator (non-smooth quadratic control case)

Require: Family type, parameter ranges, number of instances N, gain type (state-dependent/constant), homotopy parameters, seeds.

for each instance $i = 1$ to N **do**
  Sample parameter vector ψ_i from family-specific distribution $\mathcal{P}_\Psi$.
  Extract matrices (P, R), input map g, gain K from $\mathcal{P}_\Psi$..
  Compute Clarke subdifferential $\partial_C V(s)$ analytically (smooth regions and switching surfaces).
  Choose measurable selection $p(s) \in \partial_C V(s)$; on switching surfaces, fix homotopy parameter α.
  Construct drift $f(s) = -K(s)p(s)$ and optimal policy $a^*(s) = -R^{-1}g(s)^\top p(s)$.
  Compute state cost $q(s)$ via (38).
  Validate: (i) $B(s) = R + g(s)^\top P(s)g(s) \succ 0$ on grid; (ii) HJB residual $< \varepsilon$; (iii) closed-loop boundedness under π^*.
  Export fixture (YAML) with parameters, seeds, reference evaluators.
end for
return Dataset of N validated fixtures.

If $\beta \neq \gamma$, the x-component of the gradient jumps by $2(\beta - \gamma)y$. Hence, β and γ directly control the magnitude of the gradient discontinuity. The sharpness of the kink in the value function at the switching line is proportional to $|\beta - \gamma|$. These are free parameters, subject to the condition that the matrices

$$P^+ = \begin{pmatrix} 1 & \beta \\ \beta & 2 \end{pmatrix}, \qquad P^- = \begin{pmatrix} 2 & \gamma \\ \gamma & 2 \end{pmatrix}$$

must be positive definite. This requires

$$\det P^+ = 2 - \beta^2 > 0, \qquad \det P^- = 4 - \gamma^2 > 0,$$

i.e. $|\beta| < \sqrt{2}$ and $|\gamma| < 2$. They are sampled independently from any distribution supported inside these intervals.

Both pieces of V agree at $x = 0$ ($V(0, y) = 2y^2$), so V is continuous. The gradient, however, is discontinuous on the switching line $x = 0$:

$$\nabla V(x, y) = \begin{cases} (2x + 2\beta y,\ 2\beta x + 4y)^\top, & x > 0, \\ (4x + 2\gamma y,\ 2\gamma x + 4y)^\top, & x < 0. \end{cases}$$

Taking the limits $x \to 0^\pm$ gives the two limiting gradients $(2\beta y, 4y)^\top$ and $(2\gamma y, 4y)^\top$. Hence the Clarke subdifferential on the switching line is the convex hull

$$\partial_C V(0, y) = \mathrm{co}\big\{(2\beta y, 4y)^\top,\ (2\gamma y, 4y)^\top\big\}.$$

The remaining elements of the construction are fixed as in the example: $g(s) \equiv I$ (full actuation), $\rho(a) = \frac{1}{2}\|a\|^2$, and a constant gain $K(s) = kI$ with $k > 0$. Any element from this convex hull can be written as a convex combination

$$p = \alpha p^+ + (1 - \alpha p^-), \quad \alpha \in [0, 1]\,.$$

Substituting the expressions for p^+ and p^-:

$$p_\alpha(0, y) = \alpha \begin{pmatrix} 2\beta y \\ 4y \end{pmatrix} + (1 - \alpha) \begin{pmatrix} 2\gamma y \\ 4y \end{pmatrix} = \begin{pmatrix} 2\big(\alpha\beta + (1 - \alpha)\gamma\big)y \\ 4y \end{pmatrix}.$$

The parameter α is the homotopy parameter that selects one particular generalized gradient from the subdifferential. It ranges from 0 (pure left gradient) to 1 (pure right gradient), with intermediate values giving blended gradients.

In the smooth regions $x \neq 0$, the subdifferential is a singleton, so we simply take $p(s) = \nabla V(s)$:

$$p(s) = \begin{cases} (2x + 2\beta y,\ 2\beta x + 4y)^\top, & x > 0, \\ (4x + 2\gamma y,\ 2\gamma x + 4y)^\top, & x < 0. \end{cases}$$

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Notice that taking the limits $x \to 0^{\pm}$ in these smooth-region formulas recovers exactly p^+ and p^-, and the homotopy selection on the switching line interpolates continuously between them.

With this selection the construction of Subsection G.5 yields the drift, optimal policy, and state cost:

$$f(s) = -k\,p(s), \qquad a^*(s) = -p(s), \qquad q(s) = \Big(\frac{1}{2} + k\Big)\|p(s)\|^2.$$

The family is parameterised by the quadruple $\psi = (\beta, \gamma, k, \alpha)$. Every choice of these parameters produces a different control problem; all share the same optimal value function V, but the drift, optimal feedback, and state cost differ on the switching surface. The generator pipeline simply samples $(\beta, \gamma, k, \alpha)$ from user-specified distributions, computes the piecewise explicit formulas for f, a^*, q, and proceeds to the validation step described below.

G.10 Validation

Each generated instance undergoes three validation checks, applied on a grid $\mathcal{G}$ that includes points on the switching line $x = 0$.

1. **Positive definiteness of $B(s)$.** Here $B(s) = R + g(s)^\top P(s) g(s) = I + P(s)$, where $P(s)$ is the piecewise quadratic matrix of V. Both P^+ and P^- are positive definite for any β, γ (their determinants are $4 - \beta^2$ and $4 - \gamma^2$, respectively; we restrict the sampling to ensure these remain positive). Thus $B(s) \succ 0$ everywhere.
2. **HJB residual.** For each test point $s \in \mathcal{G}$, evaluate the Hamiltonian
$$H(s) = \inf_{a \in \mathbb{R}^2} \big\{ q(s) + \tfrac{1}{2}\|a\|^2 + p(s) \cdot (f(s) + g(s)a) \big\}.$$
By construction, for the selected generalised gradient $p(s)$ employed in the generator, this infimum equals zero. Numerically we verify $\max_{s \in \mathcal{G}} |H(s)| < 10^{-12}$.
3. **Boundedness.** The closed-loop system $\dot{s} = f(s) + g(s)a^*(s)$ is integrated over a fixed time horizon using a standard ODE solver. Trajectories from a set of representative initial states remain within a bounded region, confirming that the optimal policy stabilises the system and that the numerical implementation is well-behaved.

G.11 Dataset Construction

A dataset of benchmark instances is built by sampling the parameters independently from specified distributions. For instance,

$$\beta, \gamma \sim \mathcal{U}(-1.4, 1.4), \qquad k \sim \mathcal{U}(0.2, 2.0), \qquad \alpha \sim \mathcal{U}(0, 1),$$

with the additional constraint that $4 - \beta^2 > 0$ and $4 - \gamma^2 > 0$ (ensuring positive-definiteness of $P^{\pm}$). For each sampled quadruple, the piecewise expressions for f, a^*, q are computed analytically and exported as a YAML fixture, together with deterministic seeds and reference evaluators for V and a^*. The resulting collection of problems provides a testbed for reinforcement learning algorithms on a continuum of non-smooth optimal control tasks.

G.12 Remark on Higher-Dimensional Extensions

This piecewise-quadratic family example demonstrates that the non-smooth theoretical framework can be turned into a concrete benchmark generator. Extending the pipeline to higher-dimensional non-smooth value functions (e.g., piecewise quadratic with multiple switching surfaces) and to discrete-time dynamics is the subject of ongoing work.

NeurIPS Paper Checklist

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   Answer: [Yes]

   Justification: The abstract and introduction state the paper's scope: a benchmark methodology based on converse optimality, two experimental modes, and representative examples. The claims are calibrated by the limitations section and do not present the fixtures as replacements for broad RL suites or real-world validation.

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2. **Limitations.**

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   Justification: Section 7 discusses the constructed nature of the fixtures, generator overfitting risk, and open extensions including constraints, contact dynamics, partial observability, image observations, and temporally correlated disturbances.

3. **Theory assumptions and proofs.**

   Answer: [Yes]

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4. **Experimental result reproducibility.**

   Answer: [Yes]

   Justification: Appendix A specifies the reproducibility objects required to regenerate fixtures and evaluations: generator parameters, seeds, validation grids, accepted-instance metadata, CRN schedules, baseline configurations, and software versions.

5. **Open access to data and code.**

   Answer: [Yes]

   Justification: The public project page hosts the paper, code links, fixtures, experimental summaries, and visual diagnostics. For anonymous review, the page and repository should remain anonymized.

6. **Experimental setting/details.**

   Answer: [Yes]

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7. **Experiment statistical significance.**

   Answer: [Yes]

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8. **Experiments compute resources.**

   Answer: [Yes]

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9. **Code of ethics.**

   Answer: [Yes]

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